



**JAKARTA MRT CONSTRUCTION
MANAGEMENT CONSULTANTS – JICA LOAN NO. IP-554**
ORIENTAL CONSULTANTS CO., LTD JOINT VENTURE AND ASSOCIATION

Jakarta, 22 May 2017
Ref. No. CNT8/RPT/2017/05-0163

Sumitomo Corporation
SUMMITAMAS 14th Floor
Jenderal Sudirman Kavling 61-62
Jakarta Selatan 12190

Attention: **Mr. Tsuyoshi NAKAMURA**
Contractor's Representative for Sumitomo Corporation

Subject: **Reply to Resubmission of Technical Design Report for Technical Specification of TIS**

Reference : (i) Sumitomo's letter Ref. No.: SC-JMCMC-T-170124 dated 25 April 2017
(ii) JMCMC's letter Ref. No.: CNT8/RPT/2017/02-0070 dated 23 February 2017
(iii) Sumitomo's letter Ref. No.: SC-JMCMC-T-170003 dated 10 January 2017
(iv) JMCMC's letter Ref. No.: CNT8/RPT/2016/11-0194 dated 4 November 2016
(v) Sumitomo's letter Ref. No.: SC-JMCMC-T-160215 dated 15 September 2016
(vi) JMCMC's letter Ref. No.: CNT8/RPT/2016/08-0163 dated 31 August 2016
(vii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160181 dated 28 July 2016
(viii) JMCMC's letter Ref. No.: CNT8/OTH/2016/07-0143 dated 18 July 2016
(ix) Sumitomo's letter Ref. No.: SC-JMCMC-T-160143 dated 15 June 2016

Dear Sir,

Reference is made to Sumitomo's (SC) letter Ref. (i): regarding Technical Design Report – Technical Specification of TIS: SRR-RST-ELC-0008I (Rev. I).

Employer's Representative (ER) would like to issue **Notice of No Objection** with type "C" comments (**NONOC**) as shown below.

- Section 3, Table 3-1, for JIS E 5006: Title should be 'Electronic equipment for rolling stock.
- Section 5.3.6: Title should be modified.
- Section 6.4: Description of PTU function is reduced from previous edition. PTU should keep the function to read the recorded data from the equipment connect to TIS, such as VVVF, APS, BECU, DCU, ACE etc. Also TIS should have software/data install/update function to the connected equipment through the communication lines.
- Section 6.7.2: ATP system test (status check) should be included.
- Section 7: SC should update following:
 - Layout of Tc cars in Figure 7-1 is complicated.
 - Need explanation that 2 RIOs are installed on Tc due to many devices to be connected.
 - Subsection 7.1 is no longer needed.
- Section 12: Figure 12.1-1 is corresponding to Figure 6.8-1 in previous edition and to be located in section 6.8.1.
- SC should confirm consistency of the contents and wordings including abbreviations with related TDRs and CDRs.

Thank you for your attention.

Yours faithfully,


Daisuke NANOJO
The Employer's Representative, JMCMC-JV

tnk/dst

T.K.

Enclosure(s): Related letters Ref. No (i) – (ix) (9 sheets)

Cc: 1) Ms. Silvia Halim, Director of Construction, PT MRTJ
2) Mr. Arif Rahmat, Project Manager for System & Services, PT MRTJ
3) Mr. Heru Nugroho, Project Manager for Construction 1 & 2, PT MRTJ
4) Mr. Asep Solihin, Engineering Division Head, PT MRTJ
5) Head of Rolling Stock Department, PT MRTJ
6) File

Ref. No.: SC-JMCMC-T-170124
Date: 25th April, 2017



Joint Venture
Oriental Consultants Co., Ltd.,
Japan International Consultants for Transportation co., Ltd. And
Pacific Consultants co., Ltd
Jl. Johar No. 20 Menteng, Jakarta Pusat - 10350

Attention: **Mr. Daisuke Nanjo** / The Employer's Representative
Cc: PT Mass Rapid Transit Jakarta
Ms. Silvia Halim/ Director of Construction
Mr. Arif Rahmat / Project Control Manager, Project Manager for CP108

Dear Sirs,

Subject: Resubmission of Technical Design Report for
Technical Specification of TIS
Construction of Jakarta Mass Rapid Transit Project
Contract Package of Rolling Stock Section, No. CP108

Reference :

- (i) JMCMC's letter Ref. No.: CNT8/RPT/2017/02-0070 dated 23rd February 2017
- (ii) Sumitomo's letter Ref. No.: SC-JMCMC-T-170003 dated 10th January 2017
- (iii) JMCMC's letter Ref. No.: CNT8/RPT/2016/11-0194 dated 4th November 2016
- (iv) Sumitomo's letter Ref. No.: SC-JMCMC-T-160215 dated 15th September 2016
- (v) JMCMC's letter Ref. No.: CNT8/RPT/2016/08-0163 dated 31st August 2016
- (vi) Sumitomo's letter Ref. No.: SC-JMCMC-T-160181 dated 28th July 2016
- (vii) JMCMC's letter Ref. No.: CNT8/OTH/2016/07-0143 dated 18th July 2016
- (viii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160143 dated 15th June 2016

In accordance with the Employer's Requirements-General Specification 4.10 for the Contract Package No.CP108, and the above referenced document Ref. No.: (i) – (viii), we herewith resubmit the captioned document (Ref. No: SRR-RST-ELC-00081) for your review and approval.

Your kind attention to the above shall be highly appreciated.
Sincerely yours,

A handwritten signature in black ink, appearing to read 'Tsuyoshi Nakamura', written over a horizontal line.
Tsuyoshi Nakamura
Contractor's Representative
Sumitomo Corporation

Attached document;

1. Technical Design Report for Technical Specification of TIS Rev. I



JAKARTA MRT CONSTRUCTION
MANAGEMENT CONSULTANTS – JICA LOAN NO. IP-554
ORIENTAL CONSULTANTS CO., LTD JOINT VENTURE AND ASSOCIATION

Jakarta, 23 February 2017
Ref. No. CNT8/RPT/2017/02-0070

Sumitomo Corporation
SUMMITAMAS 14th Floor
Jenderal Sudirman Kaveloing 61-62
Jakarta Selatan 12190

Attention: **Mr. Tsuyoshi NAKAMURA**
Contractor's Representative for Sumitomo Corporation

Subject: **Reply to Resubmission of Technical Design Report for Technical Specification of TIS**

Reference : (i) Sumitomo's letter Ref. No.: SC-JMCMC-T-170003 dated 10 January 2017
(ii) JMCMC's letter Ref. No.: CNT8/RPT/2016/11-0194 dated 4 November 2016
(iii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160215 dated 15 September 2016
(iv) JMCMC's letter Ref. No.: CNT8/RPT/2016/08-0163 dated 31 August 2016
(v) Sumitomo's letter Ref. No.: SC-JMCMC-T-160181 dated 28 July 2016
(vi) JMCMC's letter Ref. No.: CNT8/OTH/2016/07-0143 dated 18 July 2016
(vii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160143 dated 15 June 2016

Dear Sir,

Reference is made to Sumitomo's (SC) letter Ref. (i): regarding Technical Design Report – Technical Specification of TIS: SRR-RST-ELC-0008G (Rev. G).

Employer's Representative (ER) would like to issue **Notice of No Objection** with type "C" comments (**NONOC**) as shown below.

- AMENDMENT SHEET: Listed items for this revision are not sufficient.
- Many miss descriptions of Table and Figure numbers in main statements.
- Section 5.3: Specification of 'enclosure' is not given.
- Section 6.4.2: TDR for data recorder describes data recorder uses CF card.
- Section 6.8: Screen examples should be updated reflecting the design meetings.
- Section 7, Figure 7-1: No abbreviation for EPFC.
- Section 7: Description for timer relay should be corrected.
- Section 7.1: No abbreviation for PCC.
- Section 9.2: Explanation for RIO should be updated. No enough explanation was given how 'ring topology' can be implemented by using the RIO.
- Many miss descriptions for superscript (especially in Section 10).
- Section 10.3.5: No longer needed.
- Section 12.3 and 12.4: TBD and X should be fixed.
- Section 15, table 15-1: Marking of 'X' should be changed to '✓'.
- SC should confirm consistency of the contents with the other related TDRs.
- SC should confirm corresponding CDR contains referred information. Document reference number should be added when the other document(s) would be referred.

Thank you for your attention.

Yours faithfully,

Daisuke NANJO

The Employer's Representative, JMCMC-JV

tnk/dst

T.K.

Enclosure(s): Related letters Ref. No (i) – (vii) (7 sheets)

- Cc: 1) Ms. Silvia Halim, Director of Construction, PT. MRTJ
2) Mr. Heru Nugroho, Project Manager for Construction 1 & 2, PT.MRTJ
3) Mr. Arif Rahmat, Control & Monitoring Division Head and Project Manager for System & Services, PT.MRTJ
4) Mr. Asep Solihin, Engineering Division Head, PT.MRTJ
5) Head of Rolling Stock Department, PT.MRTJ
6) File

Ref. No.: SC-JMCMC-T-170003
Date: 10th January, 2017



Sumitomo Corporation

Joint Venture
Oriental Consultants Co., Ltd.,
Japan International Consultants for Transportation co., Ltd. And
Pacific Consultants co., Ltd
Jl. Johar No. 20 Menteng, Jakarta Pusat - 10350

Attention: **Mr. Daisuke Nanjo** / The Employer's Representative
Cc: PT Mass Rapid Transit Jakarta
Ms. Silvia Halim/ Director of Construction
Mr. Arif Rahmat / Project Control Manager, Acting Project Manager for CP108
Mr. Mohamad Deddy Gamawan / Rolling Stock Department Head

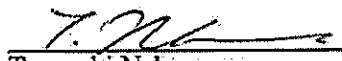
Dear Sirs,

Subject: Resubmission of Technical Design Report for
Technical Specification of TIS
Construction of Jakarta Mass Rapid Transit Project
Contract Package of Rolling Stock Section, No. CP108

Reference : (i) JMCMC's letter Ref. No.: CNT8/RPT/2016/11-0193 dated 1st November 2016
(ii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160217 dated 19th September 2016
(iii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160215 dated 16th September 2016
(iv) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0167 dated 2nd September 2016
(v) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0168 dated 2nd September 2016
(vi) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0164 dated 1st September 2016
(vii) JMCMC's letter Ref. No.: CNT8/RPT/2016/08-0163 dated 31st August 2016
(viii) JMCMC's letter Ref. No.: CNT8/RPT/2016/08-0162 dated 31st August 2016
(ix) Sumitomo's letter Ref. No.: SC-JMCMC-T-160190 dated 2nd August 2016
(x) Sumitomo's letter Ref. No.: SC-JMCMC-T-160186 dated 30th July 2016
(xi) Sumitomo's letter Ref. No.: SC-JMCMC-T-160183 dated 29th July 2016
(xii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160181 dated 28th July 2016
(xiii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160179 dated 27th July 2016
(xiv) JMCMC's letter Ref. No.: CNT8/OTH/2016/07-0143 dated 19th July 2016
(xv) Sumitomo's letter Ref. No.: SC-JMCMC-T-160143 dated 15th June 2016

In accordance with the Employer's Requirements-General Specification 4.10 for the Contract Package No.CP108, and the above referenced document Ref. No.: (i) – (xv), we herewith submit the captioned document (Ref. No: SRR-RST-ELC-0008G) for your review and approval.

Your kind attention to the above shall be highly appreciated.
Sincerely yours,


Tsuyoshi Nakamura
Contractor's Representative
Sumitomo Corporation

Attached document;

1. Technical Design Report for Technical Specification of TIS Rev. G



Jakarta, 4 November 2016
Ref. No. CNT8/RPT/2016/11-0194

Sumitomo Corporation
SUMMITMAS 14th Floor
Jl. Jenderal Sudirman Kavling 61-62
Jakarta Selatan 12190

Attention: **Mr. Tsuyoshi NAKAMURA**
Contractor's Representative

Subject: **Reply to Resubmission of Technical Design Report for Electric System**

Reference : (i) Sumitomo's letter Ref. No.: SC-JMCMC-T-160215 dated 15 September 2016
(ii) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0168 dated 2 September 2016
(iii) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0167 dated 2 September 2016
(iv) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0164 dated 1 September 2016
(v) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0163 dated 31 August 2016
(vi) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0162 dated 31 August 2016
(vii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160190 dated 1 August 2016
(viii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160186 dated 30 July 2016
(ix) Sumitomo's letter Ref. No.: SC-JMCMC-T-160183 dated 29 July 2016
(x) Sumitomo's letter Ref. No.: SC-JMCMC-T-160181 dated 28 July 2016
(xi) Sumitomo's letter Ref. No.: SC-JMCMC-T-160179 dated 27 July 2016
(xii) JMCMC's letter Ref. No.: CNT8/OTH/2016/07-0143 dated 18 July 2016
(xiii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160143 dated 15 June 2016

Dear Sir,

Reference is made to Sumitomo's (SC) letter Ref. (i), regarding Auxiliary Circuit: SRR-RST-ELC-0004G (Rev.G), Door Circuit: SRR-RST-ELC-0005C (Rev.C), TIS: SRR-RST-ELC-0008F (Rev.F), Data Recorder: SRR-RST-ELC-0009D (Rev.D), Door: SRR-RST-ELC-0010E (Rev.E), Pubric Address: SRR-RST-ELC-0011F (Rev.F), Indication Dvice: SRR-RST-ELC-0012E (Rev.E), Battery: SRR-RST-ELC-0014I (Rev.I), Lighting: SRR-RST-ELC-0015G (Rev.G), Head Light and Tail light: SRR-RST-ELC-0016G (Rev.G) and CCTV system: SRR-RST-ELC-0020D (Rev.D).

Based on the explanatory meeting between SC and JMCMC held on 1st, 3rd, 4th and 6th October 2016, Employer's Representative (ER) issues Notice of No Objection with type "B" comments (NONOB). Main points to be remarked are shown in the attached sheets the title of which is 'Employer's Representative Comments'. SC is required to revise in accordance with the comments, but not limited to those and to resubmit.

Thank you for your attention.

Yours faithfully,

Daisuke NANJO
The Employer's Representative, JMCMC-JV

lnk/dst

T.k.

Enclosure(s): 1) ER's comments to "Technical Design Report for Electric System (3 sheets)
2) Related letter Ref. No. (i)-(xiii) - (27 sheets)

Cc: 1) Ms. Silivia Halim, Director of Construction, PT. MRTJ
2) Mr. Heru Nugroho, Project Manager for Construction & Facilities, PT. MRTJ
3) Mr. Arif Rahmat, Project Management Office Division Head, Acting PM for Railway System, Trackwork & Rolling Stock, PT. MRTJ
4) Mr. Deddy Gamawan, Head of Rolling Stock Department, PT. MRTJ
5) File

Ref. No.: SC-JMCMC-T-160215
Date: 15th September, 2016

 **Sumitomo Corporation**

Joint Venture
Oriental Consultants Co., Ltd.,
Japan International Consultants for Transportation co., Ltd. And
Pacific Consultants co., Ltd
Jl. Johar No. 20 Menteng, Jakarta Pusat - 10350

Attention: **Mr. Daisuke Nanjo / The Employer's Representative**

Cc: PT Mass Rapid Transit Jakarta
Mr. Silvia Halim / Director of Construction
Mr. Arif Rahmat / Project Control Manager, Acting Project Manager for CP108
Mr. Mohamad Deddy Gamawan / Rolling Stock Department Head

Dear Sirs,

Subject: Resubmission of Technical Design Report for Electric System
Construction of Jakarta Mass Rapid Transit Project
Contract Package of Rolling Stock Section, No. CP108

Reference :

- (i) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0167 dated 2nd September 2016
- (ii) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0168 dated 2nd September 2016
- (iii) JMCMC's letter Ref. No.: CNT8/RPT/2016/09-0164 dated 1st September 2016
- (iv) JMCMC's letter Ref. No.: CNT8/RPT/2016/08-0163 dated 31st August 2016
- (v) JMCMC's letter Ref. No.: CNT8/RPT/2016/08-0162 dated 31st August 2016
- (vi) Sumitomo's letter Ref. No.: SC-JMCMC-T-160190 dated 2nd August 2016
- (vii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160186 dated 30th July 2016
- (viii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160183 dated 29th July 2016
- (ix) Sumitomo's letter Ref. No.: SC-JMCMC-T-160181 dated 28th July 2016
- (x) Sumitomo's letter Ref. No.: SC-JMCMC-T-160179 dated 27th July 2016
- (xi) JMCMC's letter Ref. No.: CNT8/OTH/2016/07-0143 dated 19th July 2016
- (xii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160143 dated 15th June 2016

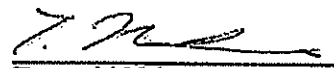
In accordance with the Employer's Requirements-General Specification 4.10 for the Contract Package No. CP108, and the above referenced letters Ref. No.: (i) – (xii) we herewith submit the captioned document with following contents for your review and approval.

Doc. No.	Rev.	Description	Status
SRR-RST-ELC-0001B	B	Symbols of Schematic diagram	NONO-C
SRR-RST-ELC-0002A	A	Allocation of Line Number	NONO-C
SRR-RST-ELC-0004G	G	Functional Explanation of Auxiliary Circuit	Attached
SRR-RST-ELC-0005C	C	Functional Explanation of Door Circuit (Including between Train Doors and Platform Screen Doors)	Attached
SRR-RST-ELC-0006C	C	Functional Explanation of Control Circuit	Preparing
SRR-RST-ELC-0007E	E	Technical Specification of Auxiliary Power Supply	Preparing
SRR-RST-ELC-0008F	F	Technical Specification of TIS	Attached
SRR-RST-ELC-0009D	D	Specification of Data Record	Attached
SRR-RST-ELC-0010E	E	Technical Specification of Door	Attached
SRR-RST-ELC-0011F	F	Specification of Public Address	Attached
SRR-RST-ELC-0012E	E	Specification of Indication Device	Attached
SRR-RST-ELC-0013E	E	Specification of Air Conditioner System	Preparing
SRR-RST-ELC-0014I (SRR-RST-ELC-0003)	I	Specification of Battery and calculation of Battery Load	Attached
SRR-RST-ELC-0015G	G	Specification of Lighting	Attached
SRR-RST-ELC-0016G	G	Specification of Head Light, and Tail Light	Attached
SRR-RST-ELC-0017D	D	Specification of Vibration Measuring Instrument	NONO-C
SRR-RST-ELC-0018E	E	Specification of Pantograph	Preparing
SRR-RST-ELC-0020D	D	Specification of CCTV System for Passenger Saloon	Attached

Attached: Included in this letter, **Preparing:** Preparing revisions, **NONO-C:** Obtained NONO-C,

Your kind attention to the above shall be highly appreciated.

Sincerely yours,


Tsuyoshi Nakamura
Contractor's Representative
Sumitomo Corporation

Attached document;
(1) Technical Design Report for Electric System



JAKARTA MRT CONSTRUCTION
MANAGEMENT CONSULTANTS – JICA LOAN NO. IP-554
ORIENTAL CONSULTANTS CO., LTD JOINT VENTURE AND ASSOCIATION

Jakarta, 31 August 2016
Ref. No. CNT8/RPT/2016/08-0163

Sumitomo Corporation
SUMMITMAS 14th Floor
Jl. Jenderal Sudirman Kavling 61-62
Jakarta Selatan 12190

Attention: Mr. Tsuyoshi NAKAMURA
Contractor's Representative

Subject: Reply to Resubmission of Technical Design Report for Electric System No.2

Reference : (i) Sumitomo's letter Ref. No.: SC-JMCMC-T-160181 dated 28 July 2016
(ii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160179 dated 27 July 2016
(iii) JMCMC's letter Ref. No.: CNT8/OTH/2016/07-0143 dated 19 July 2016
(iv) Sumitomo's letter Ref. No.: SC-JMCMC-T-160143 dated 15 June 2016

Dear Sir,

Reference is made to the Sumitomo's (SC) letter Ref. (i): regarding Door Circuit: SRR-RST-ELC-0005B (Rev.B), TIS: SRR-RST-ELC-0008D (Rev.D), Door: SRR-RST-ELC-0010C (Rev.C), Lighting: SRR-RST-ELC-0015E (Rev.E), and CCTV system: SRR-RST-ELC-0020C (Rev.C).

Based on the explanatory meeting between SC and JMCMC held on 30th July and 4th August 2016, Employer's Representative (ER) issues Notice of Objection (NOO) with type "A" comments. Main points to be remarked are shown in the attached sheets with title 'Employer's Representative Comments'. SC is required to revise in accordance with the comment, but not limited to those and to resubmit.

Thank you for your attention.

Yours faithfully,

Daisuke NANJO
The Employer's Representative, JMCMC-JV

Ink/dst
T.k

Enclosure(s): 1) ER's comments to "Technical Design Report for Electric System 2 (2 sheets)
2) Related letter Ref. No. (i)-(iv) - (8 sheets)

Cc: 1) Mr. M. Nasyir, Director of Construction, PT MRTJ
2) Mr. Heru Nugroho, Project Manager for Construction & Facilities, PT MRTJ
3) Mr. Arif Rahmat, Project Management Office Division Head, Acting PM for Railway System, Track Work & Rolling Stock, PT MRTJ
4) Mr. Deddy Gamawan, Head of Rolling Stock Department, PT MRTJ
5) File

Project Office Address :
Jl. Johar No. 20, Menteng, Jakarta Pusat 10350
Tel : 021-391 2165, 021-391 2166 Fax : 021-391 2167

Ref. No.: SC-JMCMC-T-160181
Date: 28th July, 2016

Sumitomo Corporation

Joint Venture
Oriental Consultants Co., Ltd.,
Japan International Consultants for Transportation co., Ltd. And
Pacific Consultants co., Ltd
Jl. Johar No. 20 Menteng, Jakarta Pusat - 10350

Attention: Mr. Daisuke Nanjo
The Employer's Representative

Co: PT Mass Rapid Transit Jakarta
Mr. M. Nasyir / Director of Construction
Mr. Arif Rahmat / Project Control Manager, Acting Project Manager for CP108
Mr. Mohamad Deddy Gamawan / Rolling Stock Department Head

Dear Sirs,

Subject: Resubmission of Technical Design Report for Electric System No.2
Construction of Jakarta Mass Rapid Transit Project
Contract Package of Rolling Stock Section, No. CP108

Reference : (i) Sumitomo's letter Ref. No.: SC-JMCMC-T-160179 dated 27th July 2016
(ii) JMCMC's letter Ref. No.: CNT8/OTH/2016/07-0143 dated 19th July 2016
(iii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160143 dated 15th June 2016

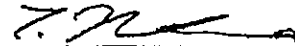
In accordance with the Employer's Requirements-General Specification 4.10 for the Contract Package No. CP108, and the above referenced letters Ref. No.: (i) - (iii) we herewith submit the captioned document with following contents for your review and approval.

Doc. No.	Rev.	Description	Status
SRR-RST-ELC-0001B	B	Symbols of Schematic diagram	Submitted
SRR-RST-ELC-0002A	A	Allocation of Line Number	Submitted
SRR-RST-ELC-0003	B	Calculation of Battery Load	Preparing
SRR-RST-ELC-0004	B	Functional Explanation of Auxiliary Circuit	Preparing
SRR-RST-ELC-0005B	B	Functional Explanation of Door Circuit (Including between Train Doors and Platform Screen Doors)	Attached
SRR-RST-ELC-0006	NIL	Functional Explanation of Control Circuit	Preparing
SRR-RST-ELC-0007	B	Technical Specification of Auxiliary Power Supply	Preparing
SRR-RST-ELC-0008D	D	Technical Specification of TIS	Attached
SRR-RST-ELC-0009	A	Specification of Data Record	Preparing
SRR-RST-ELC-0010C	C	Technical Specification of Door	Attached
SRR-RST-ELC-0011	A	Specification of Public Address	Preparing
SRR-RST-ELC-0012	NIL	Specification of Indication Device	Preparing
SRR-RST-ELC-0013	B	Specification of Air Conditioner System	Preparing
SRR-RST-ELC-0014E	E	Specification of Battery	Submitted
SRR-RST-ELC-0015E	E	Specification of Lighting	Attached
SRR-RST-ELC-0016D	D	Specification of Head Light, and Tail Light	Submitted
SRR-RST-ELC-0017D	D	Specification of Vibration Measuring Instrument	Submitted
SRR-RST-ELC-0018	B	Specification of Pantograph	Preparing
SRR-RST-ELC-0020C	C	Specification of CCTV System for Passenger Saloon	Attached

Attached: Included in this letter. Preparing: Preparing revisions. Submitted: Waiting for review result

Your kind attention to the above shall be highly appreciated.

Sincerely yours,


Tsuyoshi Nakamura
Contractor's Representative
Sumitomo Corporation

Attached document;
Technical Design Report for Electric System



**JAKARTA MRT CONSTRUCTION
MANAGEMENT CONSULTANTS – JICA LOAN NO. IP-554
ORIENTAL CONSULTANTS CO., LTD JOINT VENTURE AND ASSOCIATION**

Jakarta, 18 July 2016
Ref. No. CNT8/OTH/2016/07-0143

Sumitomo Corporation
SUMMITMAS 14th Floor
Jl. Jenderal Sudirman Kavling 61-62
Jakarta Selatan 12190

Attention: **Mr. Tsuyoshi NAKAMURA**
Contractor's Representative

Subject: **Reply to Submission of Technical Design Report**

Reference : (i) Sumitomo's letter Ref. No.: SC-JMCMC-T-160143 dated 15 June 2016

Dear Sir,

Reference is made to Sumitomo's (SC) letter Ref. (i), regarding the submitted nineteen documents with No. SSR-RST-ELC-0001 to 0018 and 0020: Electric System. Based on the explanatory meeting between SC and JMCMC held on 27 June through 16 June 2016, Employer's Representative (ER) issues Notice of NO Objection with type "C" comments (NONOC) for SSR-RST-ELC-0001 Rev.A and SSR-RST-ELC-0002 shown below:

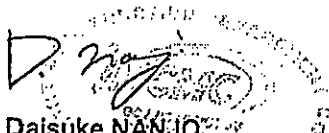
- Currently the other related technical documents are in progress to be reviewed. Contractor should apply appropriate updates with revising of the other documents if needed.

The Employer's Representative (ER) issues Notice of Objection (NOO) with type "A" comments for the documents except above mentioned (SSR-RST-ELC-0003 to 0018 and 0020). The documents did not show whether the proposing systems can satisfy the requirements of GS, PS and standards. Major comments are listed on attached 'Employer's Representative Comments'.

SC is required to revise in accordance with the above mentioned, but not limited to those and to resubmit it.

Thank you for your attention.

Yours faithfully,



Daisuke NANOJO
The Employer's Representative, JMCMC-JV

ttk/dst

7. 7.

Enclosure(s): 1) ER's comments to "Technical Design Report" (4 sheets)
2) Sumitomo's letter Ref. No. (i) (1 sheet)

Cc: 1) Mr. M. Nasyir, Director of Construction, PT MRTJ
2) Mr. Heru Nugroho, PM of Railway Systems, Trackwork and Rolling Stock, PT MRTJ
3) Mr. Arif Rahmat, Project Control & Reporting Division Head, PT MRTJ
4) Mr. Deddy Gamawan, Head of Rolling Stock Department, PT MRTJ
5) File

Project Office Address :
Jl. Johar No. 20, Menteng, Jakarta Pusat 10350
Tel : 021-391 2165, 021-391 2166 Fax : 021-391 2167

Ref. No.: SC-JMCMC-T-160143
Date: 15th June, 2016

Sumitomo Corporation

Joint Venture
Oriental Consultants Co., Ltd.,
Japan International Consultants for Transportation co., Ltd. And
Pacific Consultants co., Ltd
Jl. Johar No. 20 Menteng,
Jakarta Pusat - 10350

Attention: Mr. Daisuke Nanjo
The Employer's Representative

Cc: PT Mass Rapid Transit Jakarta
Mr. M. Nasyir / Director of Construction
Mr. Heru Nugroho / Acting Project Manager for CP108
Mr. Arif Rahmat / Project Control Manager

Dear Sirs,

Subject: Submission of Technical Design Report for Electric System
Construction of Jakarta Mass Rapid Transit Project
Contract Package of Rolling Stock Section, No. CP108

In accordance with the Employer's Requirements-General Specification 4.10 for the Contract Package No. CP108, we herewith submit the captioned document with following contents for your review and approval.

Doc. No.	Rev.	Description
SRR-RST-ELC-0001	A	Symbols of Schematic diagram
SRR-RST-ELC-0002	NIL	Allocation of Line Number
SRR-RST-ELC-0003	B	Calculation of Battery Load
SRR-RST-ELC-0004	B	Functional Explanation of Auxiliary Circuit
SRR-RST-ELC-0005	NIL	Functional Explanation of Door Circuit (Including between Train Doors and Platform Screen Doors)
SRR-RST-ELC-0006	NIL	Functional Explanation of Control Circuit
SRR-RST-ELC-0007	B	Technical Specification of Auxillary Power Supply
SRR-RST-ELC-0008	A	Technical Specification of TIS
SRR-RST-ELC-0009	A	Specification of Data Record
SRR-RST-ELC-0010	A	Technical Specification of Door
SRR-RST-ELC-0011	A	Specification of Public Address
SRR-RST-ELC-0012	NIL	Specification of Indication Device
SRR-RST-ELC-0013	B	Specification of Air Conditioner System
SRR-RST-ELC-0014	B	Specification of Battery
SRR-RST-ELC-0015	B	Specification of Lighting
SRR-RST-ELC-0016	A	Specification of Head Light, and Tail Light
SRR-RST-ELC-0017	B	Specification of Vibration Measuring Instrument
SRR-RST-ELC-0018	B	Specification of Pantograph
SRR-RST-ELC-0020	NIL	Specification of CCTV System for Passenger Saloon

Your kind attention to the above shall be highly appreciated.

Sincerely yours.


Tsuyoshi Nakamura
Contractor's Representative
Sumitomo Corporation

Attached document;
(1) Technical Design Report for Electric System

Ref. No.: SC-JMCMC-T-170124
Date: 25th April, 2017



Joint Venture
Oriental Consultants Co., Ltd.,
Japan International Consultants for Transportation co., Ltd. And
Pacific Consultants co., Ltd
Jl. Johar No. 20 Menteng, Jakarta Pusat - 10350

Attention: **Mr. Daisuke Nanjo** / The Employer's Representative
Cc: PT Mass Rapid Transit Jakarta
Ms. Silvia Halim/ Director of Construction
Mr. Arif Rahmat / Project Control Manager, Project Manager for CP108

Dear Sirs,

Subject: Resubmission of Technical Design Report for
Technical Specification of TIS
Construction of Jakarta Mass Rapid Transit Project
Contract Package of Rolling Stock Section, No. CP108

Reference :

- (i) JMCMC's letter Ref. No.: CNT8/RPT/2017/02-0070 dated 23rd February 2017
- (ii) Sumitomo's letter Ref. No.: SC-JMCMC-T-170003 dated 10th January 2017
- (iii) JMCMC's letter Ref. No.: CNT8/RPT/2016/11-0194 dated 4th November 2016
- (iv) Sumitomo's letter Ref. No.: SC-JMCMC-T-160215 dated 15th September 2016
- (v) JMCMC's letter Ref. No.: CNT8/RPT/2016/08-0163 dated 31st August 2016
- (vi) Sumitomo's letter Ref. No.: SC-JMCMC-T-160181 dated 28th July 2016
- (vii) JMCMC's letter Ref. No.: CNT8/OTH/2016/07-0143 dated 18th July 2016
- (viii) Sumitomo's letter Ref. No.: SC-JMCMC-T-160143 dated 15th June 2016

In accordance with the Employer's Requirements-General Specification 4.10 for the Contract Package No.CP108, and the above referenced document Ref. No.: (i) – (viii), we herewith resubmit the captioned document (Ref. No: SRR-RST-ELC-0008I) for your review and approval.

Your kind attention to the above shall be highly appreciated.
Sincerely yours,


Tsuyoshi Nakamura
Contractor's Representative
Sumitomo Corporation

Attached document;

1. Technical Design Report for Technical Specification of TIS Rev. I

FORM DC - DESIGN CERTIFICATE

This Design Certificate refers to the Attached Submission which comprises:

Design Package No. **SRR - RST - ELC - 0008I**

Design Stage:

- | | |
|-------------------------------------|----------------------------------|
| ☐ | Inception Design |
| ☒ | Technical Design |
| ☐ | Construction/Installation Design |
| ☐ | As-Built Document |

[description of the Works to which the submission refers]

Technical Design Report for Electric System

The contents of this submission are listed in Section A below.
Documents for which a Notice of No Objection has previously been issued and which are of relevance to this Submission are listed in Section B below.

DESIGNER'S STATEMENT

We certify that the design of the works, as shown and described in the drawings and documents listed in Section A below:

- (a) has been designed by us and complies with the Employer's Requirements and all other requirements of the Contract;
- (b) has been confirmed as a complete, adequate and valid design through an in-house design and QA check
- (c) includes all necessary and required approvals (e.g. from Local Authorities or agencies etc.) relating to the design and copies of such approvals are annexed in Section C below;
- (d) all effects of the design comprising the submission on, or from, the design of adjacent or other parts of the Works and Interfacing Contracts have been fully taken into account in the design.

[Signed by "Authorised Representative" (for Designer)]

Name: T. Michiba Date 2017/4/21
Position/Designation Chief Manager for Rolling Stock

CONTRACTOR'S CERTIFICATION

This Certifies that all design have been performed utilizing the skill and care to be expected of a professionally qualified and competent designer, experienced in work of similar nature and scope.

This further certifies that all works relating to the preparation, review, checking and certification of the design has been verified by us.

[Signed by "Authorised Representative": (for Contractor)]


 Name: Yoshitomo WATANABE Date 2017/4/21
 Position/Designation Project Manager

Section A

Submission:	☒	Drawings
	☒	Documents
	☐	Others

Please refer to the attached following "SRR" sheet:

SRR - RST - ELC - 0008I

Section B

Documents for which a Notice of No Objection has previously been issued and which are of relevance to this Submission:

Submission No.:		Date:
		Date:
Submission No.:		Date:
		Date:

Section C

[to attach copies of necessary and required approvals]

N/A

MRTJ CP108 Project

Technical Specification of Train Information System

Rev. I

15th April, 2017



Document No. SRR-RST-ELC-0008I

Submission History

Rev. Nil	; 29 th April, 2016
Rev. A	; 25 th May, 2016
Rev. B	; 20 th July, 2016
Rev. C	; 21 st July, 2016
Rev. D	; 26 th July, 2016
Rev. E	; 9 th September, 2016
Rev. F	; 12 th September, 2016
Rev. G	; 22 nd December, 2016
Rev. H	; 23 rd March, 2017
Rev. I	; 15 th April, 2017

AMENDMENT SHEET

Originator : SUMITOMO CORPORATION

DOC. Number : SRR-RST-ELC-0008I

DOC. Title : Technical Specification of Train Information System

Rev.	Date	Description	Reference Document
NIL	29 th April, 2016	First Issue	
A	25 th May, 2016	Page2: Added some Abbreviations in Table2-1	
		Page6: Added IEC 62236-3-2 in Table3	
		Page7: Added the explanation of EMI	
		Page8: Added an example of display	
		Page8: Added an explanation of the PTU	
		Page9: Added section 5.1. Supporting Functions of Maintenance	
		Page9: Added section 5.2. Passenger Service Functions	
		Page9: Added section 5.3. Operation Support Functions	
		Page11: Added Table 6	
		Page46: Added the explanation in Chapter 10. Experiences	
B	20 th July 2016	Overall Change according to internal check.	
C	21 st July 2016	Page9: Revised Table 3-1 Detailed Standards and Codes.	
D	26 th July 2016	Page9,10: Revised Table 3-1 Detailed Standards and Codes.	
		Page15,16: Added the explanation section 5.3.	
		Page20: Added the explanation section 6.3	
		Page36: Revised Table 7-1.	
E	9 th September 2016	Page 31: Added explanations in section 6.7.	
		Page 34: Modified the Figure 6.8-1.	
		Page 34 to 40: Added subsections, 6.8.1 to 6.8.7 in Section 6.8.	
		Page 54: Modified Table 9.3-1.	
F	12 th September, 2016	Page 12, Section 4-1 P.S.16.2 (b): Changed "Section 6.8.3" to "Section 6.8.4" in the column of "TDR Chapter Number".	
		Page 37: Modified the explanation in Section 6.8.4.	
G	22 nd December, 2016	Overall Change according to internal check.	
H	23 rd March, 2017	Section 4: Modified the Table 4-1.	
		Section 5: Modified the Section 5.3	

AMENDMENT SHEET

Rev.	Date	Description	Reference Document
		Section 6: Revised the Figure 6.1.2-1 and Table 6.2.2-1. Section 6: Added the Section 6.2.5. Section 6: Revised the Section 6.4, 6.5, 6.6, 6.7 and 6.8. Section 7: Revised the Figure 7-1 and added the Figure 7-2. Section 10: Deleted the Section 10.2 and 10.3. Section 11: Deleted this Section. Section 12: Moved to Section 11 and Revised this section. Section 12: Moved from CDR for TIS (SRR-RST-ELC-0024) to this section. Section 13: Moved to Section 11. Section 14: Moved to CDR for TIS (SRR-RST-ELC-0024).	
I	15 th April, 2017	Section 6.4.2, 6.6, 6.7.1, 6.7.2, 7.1, and 9.3: Modified sentences.	

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Check Sheet of Technical Design Report

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Reply Sheet from Employer’s Representative Comments

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1. Introduction

This Technical Design Report is submitted according to the requirement of the Employer's Requirement.

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2. Abbreviations and References

2.1. Abbreviations

Table 2.1-1 shows the list of abbreviations used in this report.

Table 2.1-1 List of Abbreviations

Abbreviation	Meaning
AC	Alternating Current
ACE	Air Conditioning Equipment
APS	Auxiliary Power Supply
AS	Air Suspension
ATO	Automatic Train Operation
ATP	Automatic Train Protection
BECU	Brake Electronic Control Unit
Bt	Battery
CC	Condition of Contract
CCN	CAN Open Consist Network
CCP	Communication Control Panel
CCTV	Closed-circuit Television
CCU/MON	Central Control Unit / Monitoring Unit
CDR	Construction Design Review
CP	Air Compressor
CPU	Central Processing Unit
CS	Consist Switch
CTC	Cab to Cab
DC	Direct Current
DCU	Door Control Unit
DDU	Driver's Display Unit
deg. C	degree Celsius
DGR	MINISTRY OF TRANSPORTATION Directorate General of Railways
DI	Digital Input
DO	Digital Output
ECN	Ethernet Consist Network

Abbreviation	Meaning
ED	End Device
EDD	End Destination Display
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
ESW	Ethernet Switch
ETB	Ethernet Train Backbone
ETH	Ethernet Interface
EXT	Extended Power Supply System
FIT	Failure in Time
FRP	Fiberglass-reinforced Plastic
GS	General Specification
HDLC	High-Level Data Link Control
HV	High Voltage
HVAC	Heating, Ventilation, and Air Conditioning
ID	Identification
IEC	International Electrotechnical Commission
IGBT	Insulated Gate Bipolar Transistor
I/O	Input / Output
IP	Internet Protocol
ISO	International Organization for Standardization
JIS	Japanese Industrial Standard
LAHT	Low Alloy High Tensile Steel
LCC	Lifecycle Costing
LCD	Liquid Crystal Display
LED	Light-emitting Diode
LRUs	Line-replaceable Units
M	Motor Car
MDBF	Mean Distance Between Failures
MDT	Maintenance Demonstration Test
MLIT	Ministry of Land, Infrastructure, Transport, and Tourism in Japan
MRTJ	Mass Rapid Transit Jakarta
MS	Mode Selector
MTBF	Mean Time Between Failures
MTTR	Mean Time to Restore

Abbreviation	Meaning
MVB	Multifunction Vehicle Bus
NS	Nippon Sharyo Ltd.
OCC	Operation Control Centre
OCS	Overhead Contact System
O&M	Operation and Maintenance
OS	Operating System
OSI	Open Systems Interconnection
PA	Public Address
PC	Personal Computer
PCB	Printed Circuit Board
PCH	Ethernet Check Hardware (Ethernet abnormality detector)
PCU	Propulsion Control Unit
PEC	Passenger Emergency Communication
PEI	Passenger Emergency Intercom
PI	Pulse Input
PID	Passenger Information Display
PIDC	Passenger Information Display Controller
PIS	Passenger Information System
PLC	Programmable Logic Controller
PS	Particular Specification
PSD	Passenger Saloon Display
PTU	Portable Test Unit
PWM	Pulse-width Modulation
RDT	Reliability Demonstration Test
RIO	Remote Input/Output Unit
RMDT	Reliability and Maintenance Demonstration Tests
RH	Relative Humidity
SD	Secure Digital
SDD	Side Destination Display
TBD	To Be Discussed
SAPP	Safety Assurance Program Plan
SCA	State Council for Acceptance
SCADA	Supervisory Control and Data Acquisition

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Abbreviation	Meaning
SDD	Software Design Description
TBN	Train Backbone Node
TC	Trailer Car with Cab
TCN	Train Communication Network
TFT	Thin Film Transistor
TIA/EIA	Telecommunications Industry Association/Electronic Industries Alliance
TIS	Train Information System
TM	Traction Motor
TOR	Top-of-Rail
TS	Test Specification
VGA	Video Graphics Array
VMI	Vibration Measuring Instrument
VOBC	Vehicle On-board Controller
VVVF	Variable Voltage Variable Frequency
WTB	Wire Train Bus
485	RS-485 HDLC
485a	RS-485 Asynchronous

2.2. References

Table 2.2-1 shows the list of references for this report.

Table 2.2-1 List of References

No.	Title of Document
[1]	Rolling Stock CP-108 Vol. 3: Employer's Requirement - General Specification (GS)
[2]	Rolling Stock CP-108 Vol. 4: Employer's Requirement - Particular Specification, Design and Installation (PS)
[3]	Software Quality Assurance Plan Doc. No. SRR-RST-GEN-0027
[4]	RAMS, RAM, and Maintainability Plan Doc. No. SRR-RST-RAMS-0002

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3. Codes and Standards

Table 3-1 shows the detailed standards and codes.

Table 3-1 Detailed Standards and Codes

No.	Standard and Code No.	Standard and Code Name	The Item Applied	Remarks
1	Minister of Transportation Regulation PM175 Year 2015	Standard of Technical Specification for Normal Speed Train with Own Motor	TIS	Refer to Table 4-1.
2	JIS C 0920 (2003)	Degrees of Protection Provided by Enclosure (IP Code)	TIS	Equipment shown in "The item applied" will comply with JIS C 0920, and will be verified during testing
3	JIS E 4031 (2013)	Rolling stock equipment - Shock and vibration tests	TIS	Re: ISO 3381:2005 Equipment shown in "The item applied" will comply with JIS E 4031, and will be verified during testing
4	JIS E 5006 (2005)	Electric equipment for rolling stock	TIS	Equipment shown in "The item applied" will comply with JIS E 5006, and will be verified during testing
5	IEC 61131-1 (2003)	Programmable Controllers – Part 1: General Information	TIS	Equipment shown in "The item applied" will comply with IEC 61131-1, and will be verified during testing

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No.	Standard and Code No.	Standard and Code Name	The Item Applied	Remarks
6	IEC 61131-3 (2013)	Programmable Controllers – Part 3: Programming Languages	TIS	Equipment shown in “The item applied” will comply with IEC 61131-3, and will be verified during testing
7	IEC 61375-1 (2012)	Electronic railway equipment – Train communication network - Part 1: TCN Train Communication Network General Architecture	TIS	Equipment shown in “The item applied” will comply with IEC 61375-1, and will be verified during testing
8	IEC 61375-2 Series	Electronic railway equipment - Train communication network (TCN)	TIS	Equipment shown in “The item applied” will comply with IEC 61375-2, and will be verified during testing
9	IEC 61375-3 Series	Electronic railway equipment - Train communication network (TCN)	TIS	Equipment shown in “The item applied” will comply with IEC 61375-3, and will be verified during testing
10	IEC 62236-3-2 (2008)	Railway applications – Electromagnetic compatibility – Part 3-2: Rolling stock Apparatus	TIS	Equipment shown in “The item applied” will comply with IEC 62236-3-2, and will be verified during testing
11	IEC 62278 (2010)	Railway applications – The specification and demonstration of Reliability, Availability, Maintainability, and Safety	TIS	Equipment shown in “The item applied” will comply with IEC 62278, and will be verified during testing

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No.	Standard and Code No.	Standard and Code Name	The Item Applied	Remarks
12	IEC 62279 (2015)	Railway applications – Communication signaling and processing systems – Software for railway control and protection systems	TIS	Equipment shown in “The item applied” will comply with IEC 62279, and will be verified during testing

4. Specification Compliance Matrix

Table 4-1 is the specification compliance matrix to the clause in PS that is concerned in this report.

Table 4-1 Specification Compliance Matrix

P.S. Chapter Number	Employer's Requirement	Ministry of Transportation Regulation	Comply / Not Comply	Chapter Number
16	<i>TRAIN INFORMATION SYSTEM (TIS)</i>			
16.1	<i>An open-system microprocessor-based TIS shall perform data acquisition processing, communication and display functions required to control car and train operation, and to detect, analyze and report on train operation status and subsystem failures including but not limited to:</i> <i>(a) traction equipment;</i> <i>(b) braking equipment;</i> <i>(c) auxiliary power supply;</i> <i>(d) air compressor system;</i> <i>(e) air conditioning equipment;</i> <i>(f) suspension system;</i> <i>(g) battery and battery charging system;</i> <i>(h) saloon door system;</i> <i>(i) on-board ATP/ATO equipment;</i> <i>(j) on-board communication system;</i> <i>(k) lighting system</i> <i>(l) data recording system.</i>	(2)a. Parking brake b. Motor power in rpm for diesel train and current for electric train c. Malfunction d. Speed that is completed timepiece and recorder e. Braking Air pressure f. Electrical g. Telecommunication (PM175 Clause 27)	Comply	Section 7

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P.S. Chapter Number	Employer's Requirement	Ministry of Transportation Regulation	Comply / Not Comply	Chapter Number
16.2	<i>The functions of TIS shall include, but not be limited to;</i>	-	Comply	Section 6
	<i>(a) Self inspection before putting into service incorporated with self-diagnosing of major equipment and checking of door operation, train start and train stop.</i>	-	Comply	Section 6.7
	<i>(b) Operation status display such as train speed, OCS voltage, traction motor voltage/current, auxiliary power voltage/current, battery voltage, air pressure, etc.</i>	-	Comply	Section 6.8.5
	<i>(c) Fault display and instructions to Driver. Fault information shall be shown on an overview diagram that presents the whole train and shall be shown on an individual car basis.</i>	-	Comply	Section 6.8.6
	<i>(d) Maintenance support such as fault logging, accumulated travel distance, accumulated power consumption, etc.</i>	-	Comply	Section 6.4
	<i>(e) Passenger service such as setting destination, saloon temperature and automatic PA, etc.</i>	-	Comply	Section 6.5

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P.S. Chapter Number	Employer's Requirement	Ministry of Transportation Regulation	Comply / Not Comply	Chapter Number
	<i>(g) Operation support such as setting of train number and crew number through the ATP system, and transmitting data to wayside systems through the train radio system.</i>	-	Comply	Section 6.5 Section 6.8.1 CDR/SRR-RST-ELC-0024 [Section 7.9]
16.3	<i>The TIS shall continuously record data relevant for analysis and investigation into causes of subsystem failure and train operating status immediately prior to failure.</i>	-	Comply	Section 6.4
16.4	<i>A single point failure of any individual part of the TIS shall not result in a loss of data. The performance impact of any single point failure of the TIS shall be minimized.</i>	-	Comply	Section 10
16.5	<i>The TIS shall be immune to interference or performance degradation in the environment influenced by EMI and harmonics generated from the on-board power equipment.</i>	-	Comply	Section 5.3.4
16.6	<i>The display unit shall be use TFT color LCD technology. Commands shall be entered through touch panel.</i>	-	Comply	Section 9.3 Section 11.2.6 Section 11.2.7

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P.S. Chapter Number	Employer's Requirement	Ministry of Transportation Regulation	Comply / Not Comply	Chapter Number
16.7	<i>A download facility shall be provided to download data and information from the TIS. The Contractor shall provide any software required to enable the data exchange between TIS and a laptop computer.</i>	-	Comply	Section 6.1.3 Section 6.2.4 Section 6.4 Section 11.2.8
16.8	<i>The TIS programming shall allow for easy data entry and function changing and upgrading throughout the life of the system.</i>	-	Comply	Section 6.4 Section 6.8.3
16.9	<i>The TIS shall be well-proven technology applied in rolling stock use. The software structure shall comply with IEC 61131: Programmable Controllers and communication network shall comply with IEC 61375-1: Electric railway equipment-Train bus Part 1. Train Communication Network or equivalent.</i>	-	Comply	Section 5.3
16.10	<i>The contractor shall provide the TIS configuration schemes for review by the Employer's Representative.</i>	-	Comply	Figure 7-1 Figure 7-2

5. Summary of Train Information System

5.1. Overview of Toyo Denki Seizo K.K.

Toyo Denki Seizo K.K. is a multifaceted electrical equipment manufacturing company that designs and manufactures equipment for railway applications, including propulsion systems and train information systems (TISs).

5.2. Train Communication Network Architecture of TIS

Train communication network (TCN) architecture of the TIS is shown as follows.

- No Train Backbone
- Ethernet-based Consist Network
- RS-485 Communication for Sub-Network Level Communication.

TIS is consisted as follows.

- Central Control Unit/Monitoring Unit (CCU/MON) installed in both leading cars
- Remote Input/Output Unit (RIO) installed in each car
- Driver's Display Unit (DDU) installed on both driver's consoles.

5.3. TIS Network

5.3.1. Train Backbone

It is not required because the train consist is a fixed 6-car train, which may be expanded to a fixed 8-car train in the future.

5.3.2. Consist Network

Ethernet-based consist network complies with IEC 61375 series.

5.3.3. Sub-Network

The TIS and equipment suppliers integrate to make the best network arrangement. IEC 61375 series does not specify sub-network protocols. Toyo Denki is choosing the most common, service-proven protocol used in Japan.

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5.3.4. EMI

Interface circuits of the RIO have filter circuits against EMI and shield cables are applied for communications. Enclosure of the RIO unit is made of a metal having electrostatic shield effects. Qualification tests of the TIS are performed in accordance with IEC 62236-3-2.

5.3.5. Software Structure

Software structure is in accordance with IEC 61131 series. In the future, the Side Destination Display (SDD) will be provided to ensure that software structure is in accordance with IEC 61131 series.

5.3.6. IP Code

The RIO units of the TIS are installed into the Electric Equipment Box mounted on the underfloor. The Electric Equipment Box is in accordance with JIS C 0920.

Details of the Electric Equipment Box are shown in CDR – Electric Circuit Components (SRR-RST-ELC-0056).

5.3.7. Electronic Units

Electronic units of the TIS are designed and tested in accordance with JIS E 5006 (2005). In the future, a test procedure will be provided to ensure that those units are in accordance with JIS E 5006 (2005).

5.3.8. Quality Assurance

The TIS and its software are designed, manufactured, and tested in accordance with IEC 62278 and IEC 62279. Refer to **[3]** and **[4]** for further details.

6. TIS Functions

6.1. Summary of TIS Functions

The TIS performs the following functions:

- TIS transmits the powering/braking commands and the control signals for on-board equipment.
- Manages the status of on-board equipment.
- TIS indicates the information display on the DDU.
- TIS indicates failure information from on-board equipment.
- TIS has operator support functions to view and manage failure records, operation records, and other functions.
- TIS can connect with the Portable Test Unit (PTU) to download and examine failure and operation records.

Details of the TIS functions are shown in CDR for TIS (SRR-RST-ELC-0024).

6.1.1. Indications on Monitor Display

The TIS indicates status information received from on-board equipment on the monitor display in both English and Indonesian.

When ATO/ATP is active, the TIS indicates the train speed from the ATO/ATP. If the TIS cannot obtain the information from the ATO/ATP, the TIS indicates the speed information from the BECU after adjusting for the wheel diameter.

During the mock-up review, a sample of the TIS display will be shown to the MRTJ.

6.1.2. Example of Monitor Display

The monitor display indicates failure information. The monitor display indicates a train overview diagram for the entire trainset, as well as a table of status values for individual cars.

Figure 6.1.2-1 shows an example screen of the failure screen under the ATO mode.

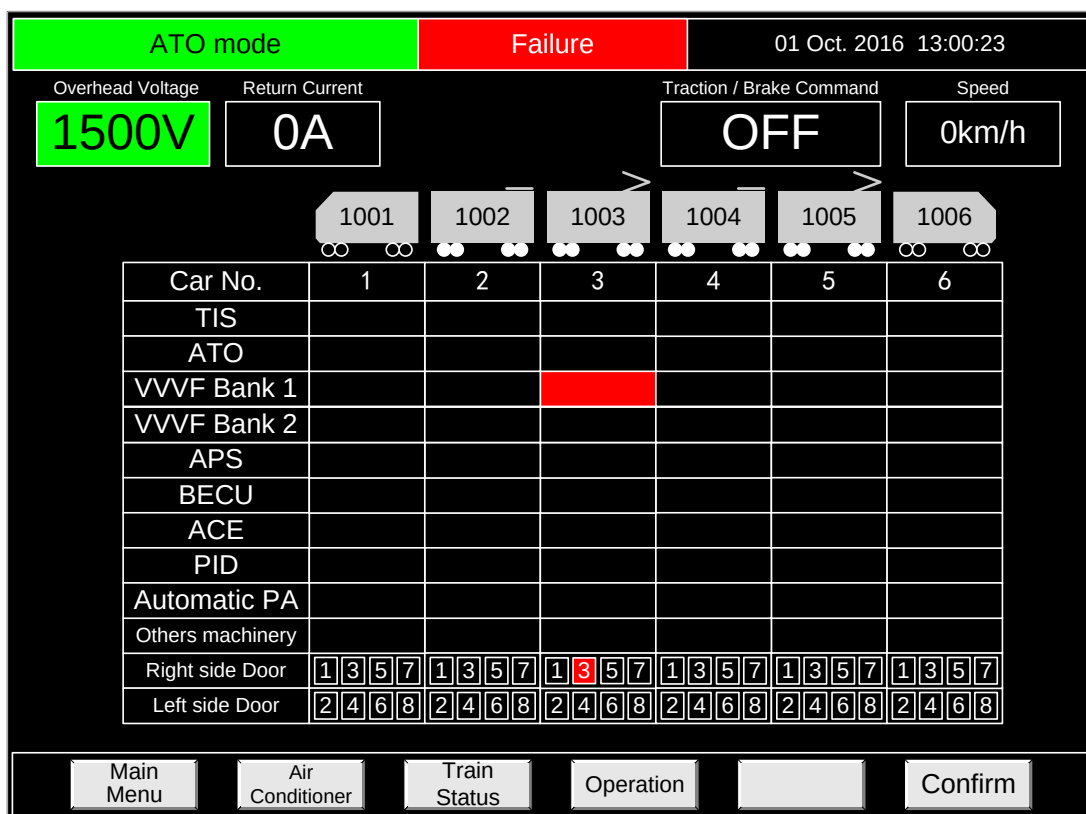


Figure 6.1.2-1 Example Display of Failure Screen

6.1.3. Management of Failure Records and Operation Records

Operators and maintenance personnel can connect the TIS to a PTU to view the management failure records and operation records. The PTU can connect to the display control unit of the DDU and then download and analyze the failure records and the operation records.

The software required to enable the data exchange between the TIS and the PTU is provided.

6.2. Basic Functions

6.2.1. Setting of Train Status

Network addresses are required to allow Ethernet communication between components of the TIS. Section 7 describes the TIS hardware configuration.

6.2.2. Train Formation Supported by TIS

Table 6.2.2-1 shows the train formation supported by the TIS.

The train formation will be expanded to 8-car trains in the future.

Table 6.2.2-1 Train Formation Supported by TIS

Car No.	6	5	4	3	2	1
Car Type	Tc2	M1	M2	M1'	M2'	Tc1

6.2.3. Indication of Train Formation

The monitor display indicates the train formation according to memorized car numbers. The TIS determines the train direction indicated on the monitor display according to the state of key switches of both leading cars. The right and left side of the screen correspond to the train running direction. The monitor display indicates the same indication both for the front cab and end cab.

Figure 6.2.3-1 shows the 6-car train formation indicated on the monitor display.

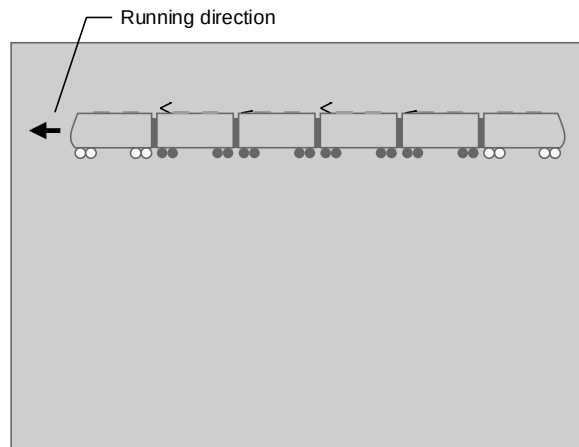


Figure 6.2.3-1 Indication of Train Formation

6.2.4. Passenger Load Factor

The monitor display and the PTU indicate the passenger load factor.

6.2.5. Auxiliary Power Extension

When an Auxiliary Power Supply (APS) failure occurs, the TIS receives a command from the APS to perform the power extension operation. The TIS then outputs a load-shed command to the air conditioners and drives the extension supply contactor.

Figure 6.2.5-1 shows the auxiliary power extension function.

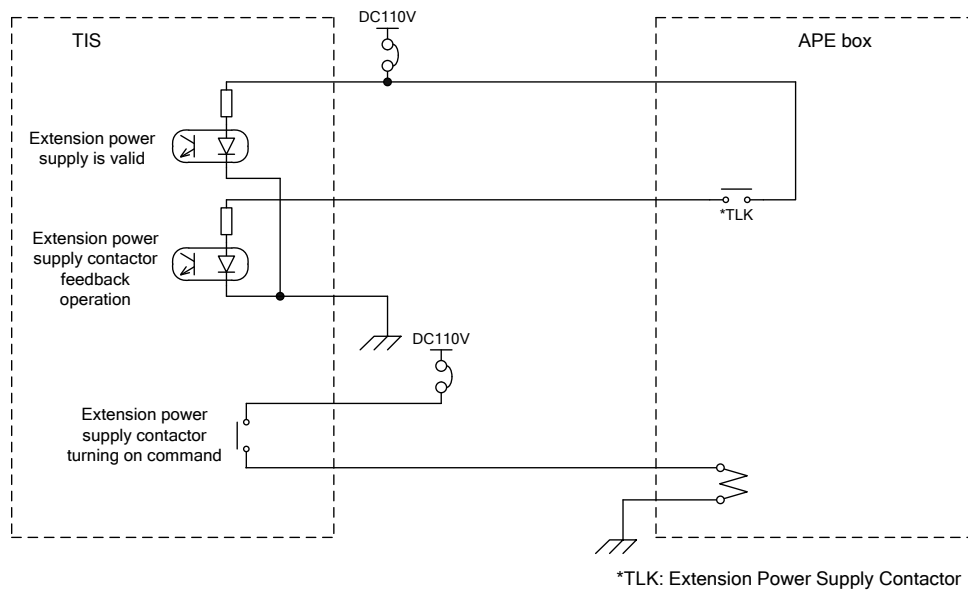


Figure 6.2.5-1 Auxiliary Power Extension Function

6.3. Powering and Braking Control Transmission Functions

The TIS transmits control commands to the Variable Voltage Variable Frequency (VVVF) inverter and the Brake Electronic Control Unit (BECU). Use of the TIS for this purpose reduces the number of dedicated trainlines and provides better functionality and monitoring, compared to the traditional hard-wired trainline arrangements.

6.4. Supporting Functions for Maintenance

Supporting functions for maintenance provide information to improve the efficiency of maintenance work. The monitor display indicates collected operating states of each device on the train. The TIS has functions to judge test results by comparing measured values with reference values.

A PTU connected to the Display Control Unit (DCU) can read out failure records, etc.

Authorized maintenance personnel can use the monitor display to modify the TIS parameters for passenger load factor. If a TIS software update is required, the personnel can use the PTU to install the software update.

6.4.1. Indication for Failure History

When a failure occurs in a car, the TIS records the Car ID, the kind of failure, and the state of the car at the time the failure occurred. The monitor display indicates the failure history for each car.

6.4.2. Data Recording and Analysis

A Data Recorder is installed in each leading car and is connected to the CCU/MON unit of the same car via RS-485 interface. Each CCU/MON unit collects and edits the operation information and sends it to the Data Recorder periodically.

The TIS can transmit operation data, failure data, and other train data for 14 days at 19 hours per day (05:00 ~ 24:00), which is the daily operating hours for MRTJ. The sampling period is 200ms. The data is stored on the CF card of the data recorders.

Maintenance personnel can use the PTU to read and analyze the events recorded to the data recorder. The PTU can also be used to make a run curve from the recorded event data.

The TIS transmits the following information to the data recorder.

- Basic information
 - o Date and time
 - o Train speed
 - o Location (distance from a station which the train departed just before).
- Information about driver's operation
 - o Status of operating control devices
 - o Status of operating service brakes
 - o Failure information.
- Behavior of ATO / ATP
 - o ATO / ATP signals received
 - o Brake commands from ATP (service, emergency)
 - o ATO commands (powering notch, braking notch).

6.4.3. Record of Running Data

The TIS can record accumulated running distance data. The monitor display in the DDU indicates accumulated running distance data on inspection screens.

6.4.4. Passenger Load Calculation

The TIS calculates the passenger load based on air suspension (AS) pressure from the BECU and from parameter values of the passenger load factor on each car. When the train departs from a station (at the time when passenger doors are closed), the TIS fixes the passenger load data in the RIO unit until the train stops at the next station to prevent from fluctuation of the passenger load data.

6.4.5. Record of Accumulated Energy Consumption Data

The TIS records accumulated energy consumption with resolution of 1 kWh. The DDU indicates the accumulated data on inspection screens. Accumulated energy consumption data are also recorded to the data recorder.

6.5. Service Device Control Functions

Service device control functions control service devices, such as air conditioning equipment (ACE) and lighting equipment. The TIS supports the device control functions described in the following subsections.

The service device control functions are:

- ACE Control
- ACE Load Shed Mode Command
- Passenger Information System (PIS) Control
- PA System Control
- Auxiliary Power Extension Feeding Operation Command
- Output to On-Board CCTV
- Output to Radio Communication System
- Lighting Control
- Detection of Battery Operation
- Detection of Battery Charger Operation

- Passenger Emergency Communication (PEC) Operation
- Detection of Air Compressor Operation.

6.6. Supporting Functions for Crew

Supporting functions for the crew display information on the DDU to inform the operator about the operating state of various devices in each car of the train.

Supporting functions for the crew are shown as following.

- Setting of information for operational use
- Indication for Operating State
- Indication for State of Car
- Failure Indication
- Indication of Propulsion System Failure
- Indication of TIS Failure
- Pantograph Operation.

6.7. Testing Functions

6.7.1. Supporting Inspection

The TIS includes functions to support self-diagnosis, monthly inspections, and test running, etc. The testing functions for supporting inspection are shown as follows.

- Self-diagnostic Functions
- Record of Measuring Information
- Record of Information for Test Running
- Basic Setting
- Collection of Record for Operation Information
- Clearance of Record for Operation Information
- Indication by the Table of Failure Record
- Collection of Failure Record
- Clearance of Failure Record of Devices
- Setting of Clock
- Interlocking Function by Password

- Collection of Record for the Test Results
- Record Collection of Self-diagnosis
- VVVF Sequence Test
- PIS Indication Test
- Simulated Running Test
- On-board Test.

6.7.2. Start-up Test

The TIS includes functions to support operators. The testing functions for start-up test are shown as follows.

- Pantograph Operation Test
- Brake Test
- Door Opening and Closing Test
- VVVF Test
- PA Test

6.8. Screen Configuration

6.8.1. Summary of Screen Configuration

The TIS indicates the information for the operator on the monitor display.

After the TIS is energized, the Main Menu display appears. The Initial Setting display can be used to manually set a train number and stations when that information is not available from the ATO/ATP. The Operation Display consists of three (3) displays: Operation, Train Status, and Failure. The Operation Display indicates a train configuration, operation commands, train speed, door operation status, etc. The Train Status display indicates status of on-board equipment. The Failure display indicates on-board equipment failures and what kind of failure occurred. The Air Condition display indicates the current and target temperatures for the passenger area.

6.8.2. Main Menu Screen

The monitor display shows the Main Menu screen. It indicates a screen name, operation mode, date and time, train configuration, and buttons to select other displays. The screen name, operation mode, and date and time commonly appear on all screens.

The Main Menu screen shows the train configuration and indicates the pantograph status.

There are eleven (11) select buttons on the Main Menu screen. When an operator touches a button, the DDU changes to the display for the selected function.

Refer to CDR for TIS (SRR-RST-ELC-0024) Attachment 3 for an example of the Main Menu screen.

6.8.3. Initial Settings Screen

When the “Initial Settings” button is touched on the Main Menu display, the DDU changes to the Initial Settings screen. On this display, an operator can manually set a train ID and starting and terminal stations. The ATO/ATP provides this information, and the operator normally does not need to manually set it.

When an operator enters new data, the operator needs to touch the “Confirm” button to apply the changes. If the Confirm button is not touched, the data is not updated with the entered data.

Refer to CDR for TIS (SRR-RST-ELC-0024) Attachment 3 for an example of the initial settings screen.

6.8.4. Operation Screen

When the “Operation” button is touched on the Main Menu screen, the DDU changes to the Operation screen. This screen indicates the train ID and station information. The station information includes a terminal station, a next station and a current station, at which the train stops or from which the train departed.

The Operation screen also indicates the train configuration and the current status to be checked by an operator during service operations. Passenger doors that are completely closed indicate in green. Doors that are not completely closed indicate in yellow.

If any on-board equipment failures occur, the car equipped with the failed equipment is indicated with a failure code shown in red.

Refer to CDR for TIS (SRR-RST-ELC-0024) Attachment 3 for an example of the operation screen.

6.8.5. Train Status Screen

When the “Train Status” button is touched on the Main Menu screen or the Operation screen, the DDU changes to the Train Status Screen. This screen indicates the current status of on-board equipment. If any on-board equipment failures occur, the car equipped with the failed equipment is indicated with a failure code shown in red.

The Train Status screen also indicates overhead voltage, line current (return current), operation commands, train speed, traction motor current, auxiliary power voltage/current, brake cock status, parking brake status, battery status, compressor operation status, and air pressure status. (Complies with PS 16.2 (b)).

Refer to CDR for TIS (SRR-RST-ELC-0024) Attachment 3 for an example of the train status screen.

6.8.6. Failure Screen

When the “Failure” button is touched on the Main Menu screen of the Operation screen, the DDU changes to the Failure screen. This screen shows red failure indicators for failures reported by on-board equipment. When a failure indicator is touched, the screen shows the failure list and failure details about the failure name, failed equipment, and guidance for the operator regarding the failure.

Refer to CDR for TIS (SRR-RST-ELC-0024) Attachment 3 for an example of the failure screen.

6.8.7. Air Conditioner Screen

When the “Air Conditioner” button is touched on the Main Menu screen, the screen will proceed to the Air Conditioner screen. The operator can set the passenger cabin temperature, operation mode, and liner fan for the air conditioners. Additionally, the screen will indicate red to show an air conditioner failure and allows the operator to reset the failure.

Refer to CDR for TIS (SRR-RST-ELC-0024) Attachment 3 for an example of the air conditioner screen.

6.8.8. Inspection Menu Screen

When the “Inspection Menu” screen is touched on the Main Menu screen, the DDU will change to the Inspection Menu screen. The screen shows any buttons that maintenance personnel can touch to select a maintenance function.

Refer to CDR for TIS (SRR-RST-ELC-0024) Attachment 3 for an example of the inspection menu screen.

7. System Configuration of TIS

The system configuration of the TIS is shown as follows.

- CCU/MON units installed in both leading cars.
- RIO units installed in each car
- DDU installed in both operator cabs.
- Ethernet cables which connect these units.
- RIO unit has a ring topology and two systems with a high level of redundancy.
- Data Recorder is connected to the CCU/MON unit in both operator cabs.
- PTU can connect to the display control unit in the DDU.

Ethernet interfaces among the CCU/MON and RIO units constitute a ring topology network.

RIO units have several kinds of interfaces to connect other devices, such as the VVVF, APS, BECU, ACE, and other equipment.

Figure 7-1 and Figure 7-2 show the system configuration and the equipment connection of the TIS.

Figure 7-1 shows the system configuration of the TIS.

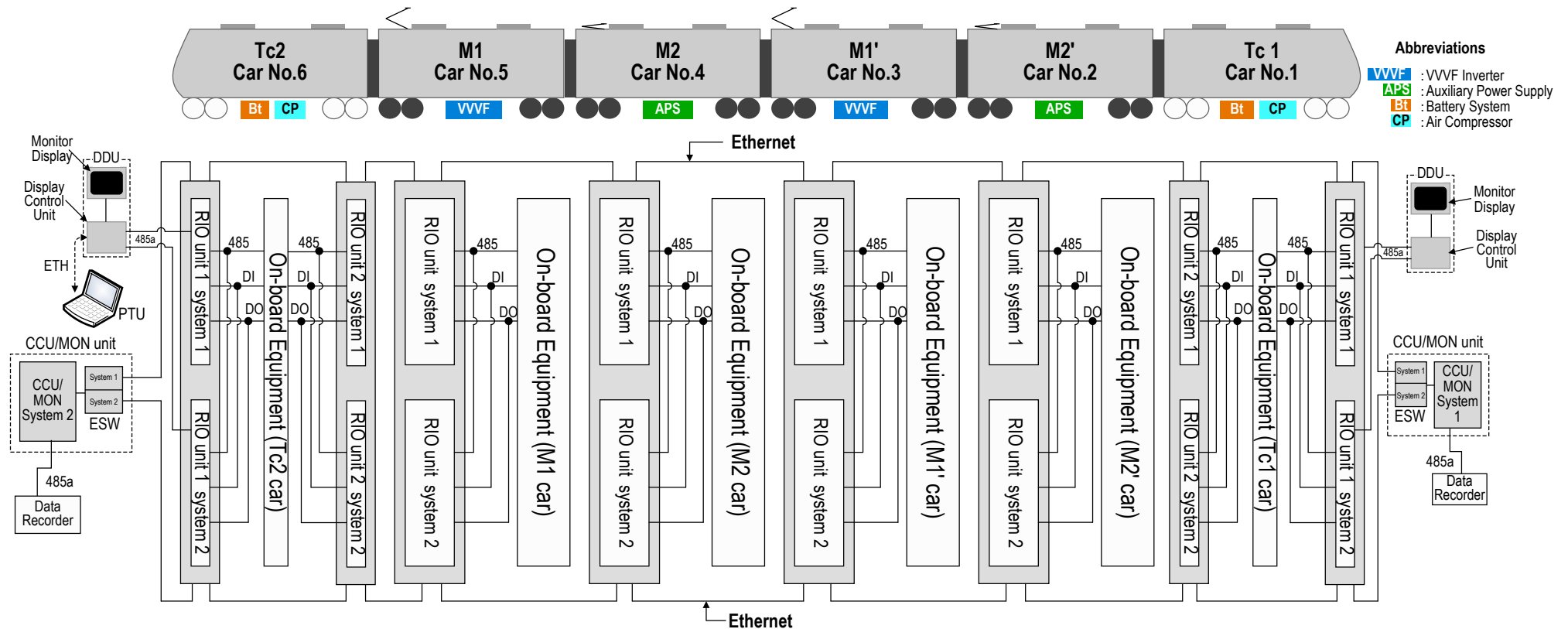


Figure 7-1 System Configuration of TIS

Details of the equipment connected to the RIO units are shown in Table 7-2.

(Filename): TDR for TIS (Rev.I).docx	Rev. I
Technical Specification of TIS	Page 32

Figure 7-2 shows the detail of equipment connections to the RIO Unit.

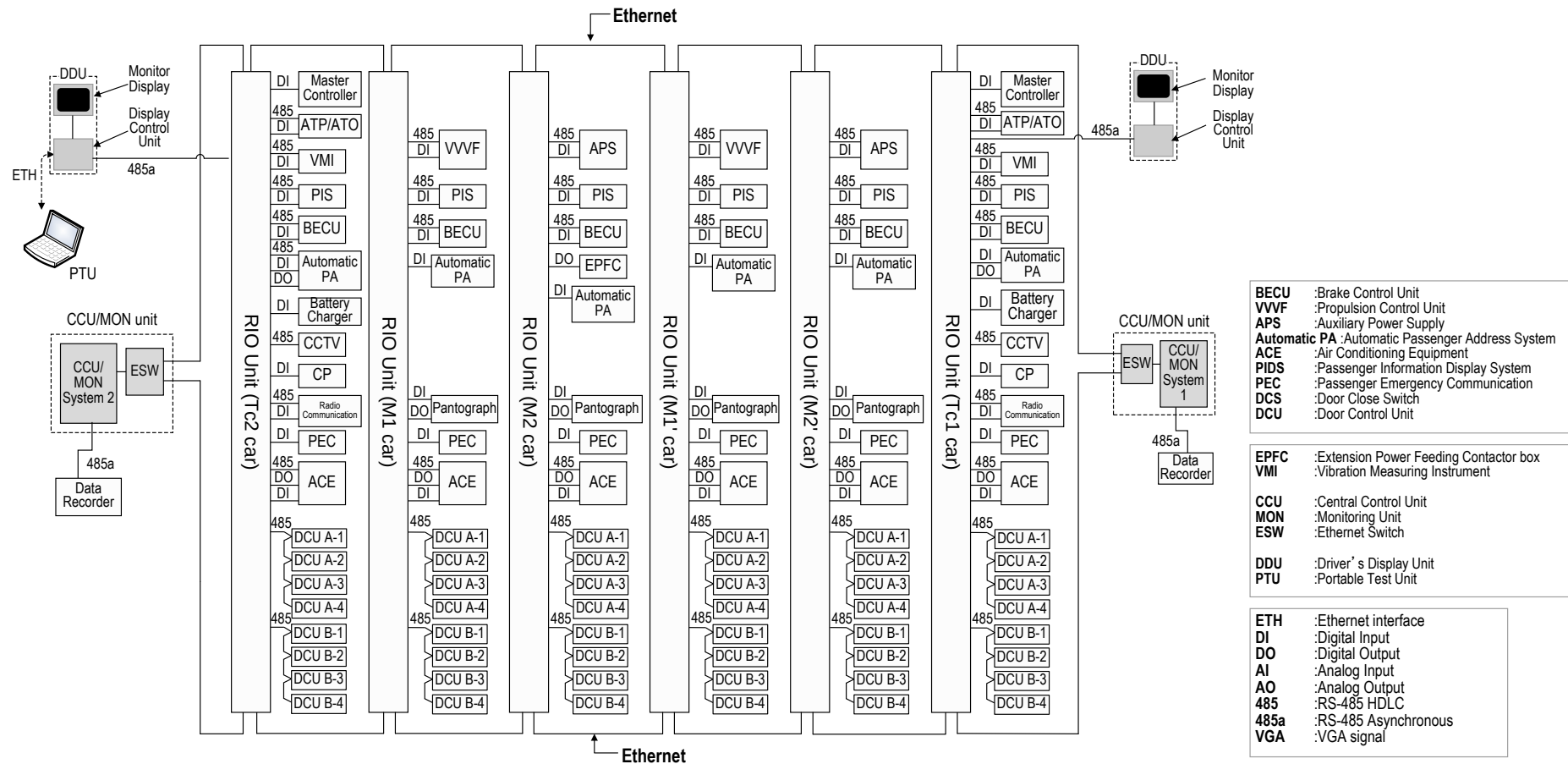


Figure 7-2 Detail of Equipment Connections to RIO Unit

Note: The air compressor system is controlled by a timer relay to prevent all of the air compressors in the train from starting at the same time. In addition, the air conditioning system has a time lag control function to prevent simultaneous activation of air conditioning compressors in the trainset. In addition, the signal requesting activation to the air compressors is input to the air conditioning system by hard-wires so that the air compressors and the air conditioning compressors can't be activated at the same time. Therefore, a huge inrush of current to the APS is prevented without relying on TIS control.

Table 7-1 shows the list of subsystems connected with the TIS.

Table 7-1 List of Subsystems Connected with TIS

Abbreviation	Subsystem	Interface	Car
VVVF	VVVF Inverter	RIO	M1 and M1'
BECU	Brake Electronic Control Unit	RIO	All Cars
APS	Auxiliary Power Supply	RIO	M2 and M2'
CP	Air Compressor	RIO	Tc1 and Tc2
ACE	Air Conditioning Equipment	RIO	All Cars
-	Battery Charger	RIO	Tc1 and Tc2
-	Battery	RIO	Tc1 and Tc2
DCU	Door Control Unit	RIO	All Cars
ATP/ATO	Automatic Train Protection/Operation	RIO	Tc1 and Tc2
PIS	Passenger Information System	RIO	Tc1 and Tc2
-	Automatic Passenger Address (PA) System	RIO	Tc1 and Tc2
PEC	Passenger Emergency Communication	RIO	All Cars
-	Radio Communication	RIO	Tc1 and Tc2
-	Data Recorder	CCU/MON	Tc1 and Tc2
CCTV	Closed-circuit Television	RIO	Tc1 and Tc2
-	Pantograph	RIO	M1, M2, M1', and M2'
EPFC	Extension Power Feeding Contactor Box	DO	M2
VMI	Vibration Measuring Instrument (Option)	RIO	Tc1 and Tc2
-	Lighting System	RIO	All Cars

Notes:

- VMI will be not provided on entire fleet.
- For equipment controlled by trainline signals, the trainline signals have higher priority than the command signals sent to the equipment by the TIS.

- The TIS is not directly connected with the suspension system because it has no communication interface with it. The TIS receives suspension system data via the BECU.

7.1. Control and Monitoring Transmission System

The CCU/MON System performs the following functions:

- Transmit powering/braking commands as described in Section 6.12.2 of the TIS CDR (SRR-RST-ELC-0024).
- Transmit control signals for devices, such as lighting and air conditioning.
- Monitor the status of the various devices as described in Section 6.12.4.2 and 6.12.4.3 of the TIS CDR (SRR-RST-ELC-0024).

CCU/MON System 1 and CCU/MON System 2 are configured as a redundant pair such that both units execute the same calculations. One of two units operates in active mode and the other operates in standby mode.

The RIO units use the output signals from the active CCU/MON unit, not from the standby CCU/MON unit. If the active CCU/MON unit fails, each RIO unit instantly changes its source of output signals from the active CCU/MON unit to the standby CCU/MON unit. This failover mechanism allows the TIS to continue operating without failure. If both the active and standby CCU/MON units fail, the TIS illuminates an alarm lamp on the driver's console and sends an abnormal status to the Propulsion Control Units (PCUs) and BECUs via hard-wired trainlines. The train can still move because the trainlines are able send control signals from the master controller and ATO/ATP to the PCUs and BECUs, even if the TIS has failed.

Figure 7.1-1 shows the basic configuration of the TIS and consists of the following units.

CCU/MON Unit

- Central Control Unit (CCU)
- Monitoring Unit (MON)
- Ethernet Switch (ESW)

RIO Unit

DDU

- Monitor Display
- Display Control Unit

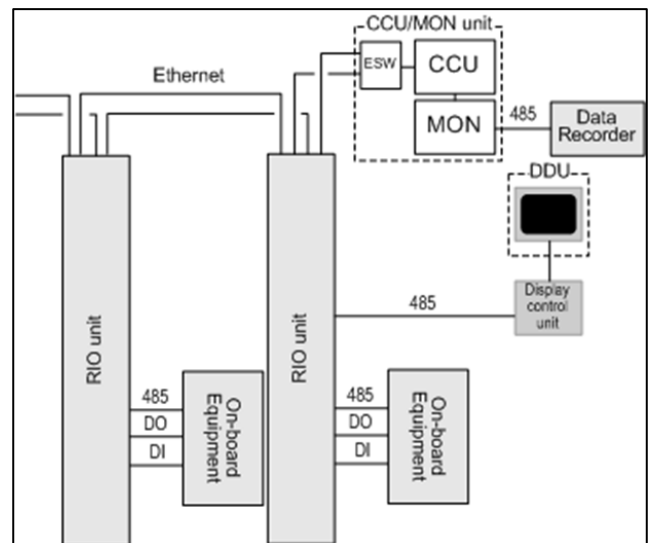


Figure 7.1-1 Basic Configuration of TIS

The TIS network is an Ethernet-based consist network and several sub-networks connecting RIO units and on-board equipment.

The failure rate of the CCU/MON unit is approx. 6.9×10^{-7} (MTBF: approx. 1.45×10^6 hours) and the probability of both CCU/MON units failing simultaneously is estimated as approx. 4.7×10^{-11} . The rate is approximately 1/1000 of the failure rate of the Ethernet Consist Network (ECN) described in Section 10.1.

Therefore, the probability of both CCU/MON units failing simultaneously has almost no influence on the probability that the TIS is completely failed.

8. Network Architecture

PS 16.9 requires that the TCN shall be in accordance with IEC 61375-1. To demonstrate that the proposed TIS satisfies the requirement, IEC 61375 series needs to be explained.

Figure 8-1 shows that the set of IEC 61375 series documents are divided into three parts.

No.	Title	Rev.	Situation	Date	
61375-1	TCN General Architecture	ed3.0	PPUB	2012-06	← Part 1
61375-2-1	WTB - Wire Train Bus	ed1.0	PPUB	2012-06	
61375-2-2	WTB Conformance Test	ed1.0	PPUB	2012-06	Part 2
61375-2-3	Communication Profile	ed1.0	PPUB	2015-07	
61375-2-4	Application Profile		CDTS		
61375-2-5	ETB - Ethernet Train Backbone	ed1.0	PPUB	2014-08	
61375-2-6	On-board to Ground Communication		ANW		
61375-3-1	MVB - Multifunction Vehicle Bus	ed1.0	PPUB	2012-06	Part 3
61375-3-2	MVB Conformance Test	ed1.0	PPUB	2012-06	
61375-3-3	CANopen Consist Network	ed1.0	PPUB	2012-06	
61375-3-4	ECN - Ethernet Consist Network	ed1.0	PPUB	2014-03	

CDTS	Circulated Draft Technical Specification
ANW	Approved New Work
PPUB	Publication issued

Figure 8-1 Configuration of IEC 61375 Series

Part 1, or IEC 61375-1 describes general architecture of the TCN. In this part, the overview of the variations of train backbones and consist networks are described. The Wire Train Bus (WTB) and the Ethernet Train Backbone (ETB) are defined as train backbones. Multifunction Vehicle Bus (MVB), CAN Open Consist Network (CCN), and ECN are defined as consist networks.

Part 2, or IEC 61375-2-1, 2-2, 2-3, 2-4, 2-5 and 2-6, describe the details of train backbones and their protocols.

Part 3, or IEC 61375-3-1, 3-2, 3-3 and 3-4 describe the details of consist networks and their protocols. Details of the ECN are described in IEC 61375-3-4.

Figure 8-2 shows the network architecture defined by IEC 61375-1.

IEC 61375-1 defines the network architecture hierarchically.

IEC 61375-1 defines the network architecture hierarchically as the train backbone level and the consist network level.

In some cases, the consist network connects to the sub-networks via gateways. Figure 8-2 shows the concept of such network architecture.

Section 6.3.3 of IEC 61375-1 explains the idea of the sub-network briefly but it doesn't describe the concrete requirements about the sub-network.

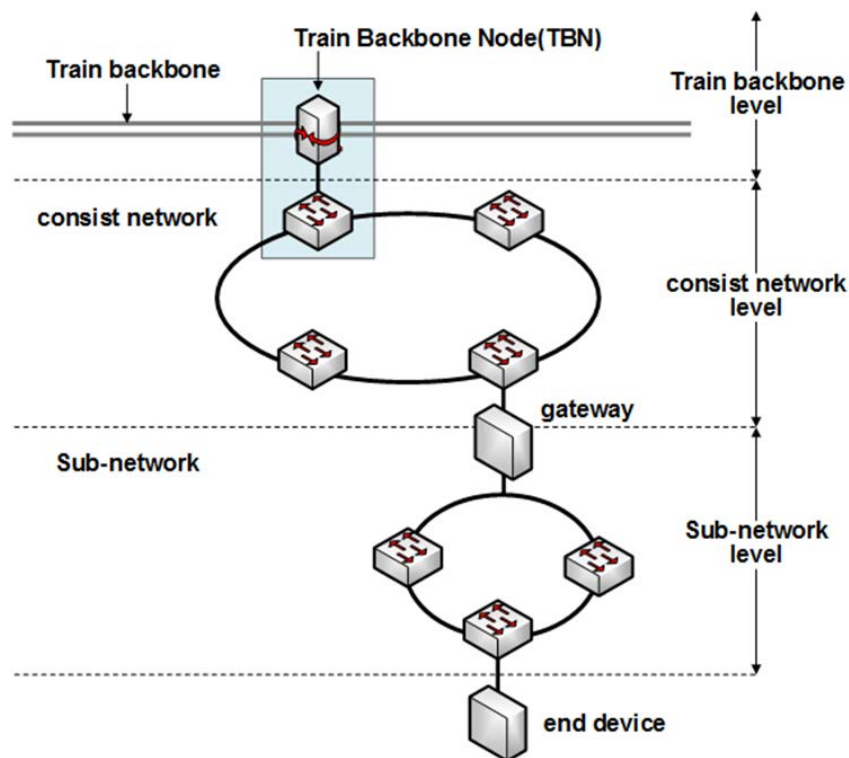


Figure 8-2 Network Architecture Defined by IEC 61375-1

The MRTJ trains are configured as 6-car consists and their formation does not change dynamically in normal operation. Therefore, the TIS for MRTJ vehicles are “closed trains” from the IEC 61375 point of view.

Figure 8-3, which shows two train consists, demonstrates the general configuration of the TCN. The train backbone was designed to implement the data exchange between multiple, different consist networks. The train backbone is necessary for trains which change their formation dynamically, but it is not necessary for closed trains.

Figure 8-3 shows the train backbone and consist networks.

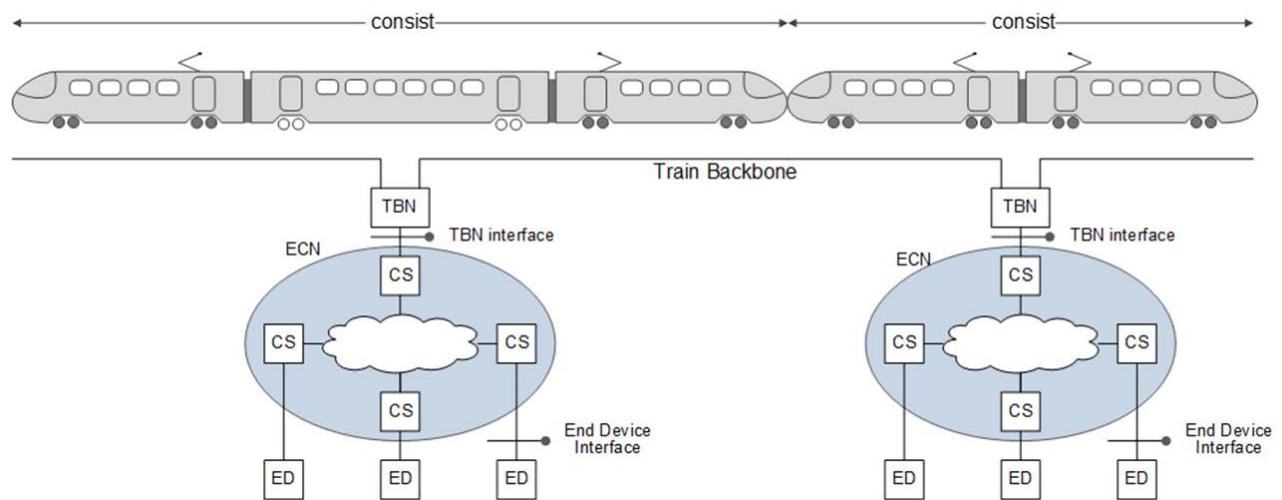


Figure 8-3 Train Backbone and Consist Networks

The TIS is one consist network and it has neither a train backbone nor a train backbone node (TBN) as shown in Figure 8-4. IEC 61375-1 Section 4.3.2 states that a consist can have zero (0) TBNs.

TBNs will not be used if a train rescues a failed train. In that situation, a hard-wire will be used to control the emergency brake.

Figure 8-4 shows the train backbone and consist networks for MRTJ.

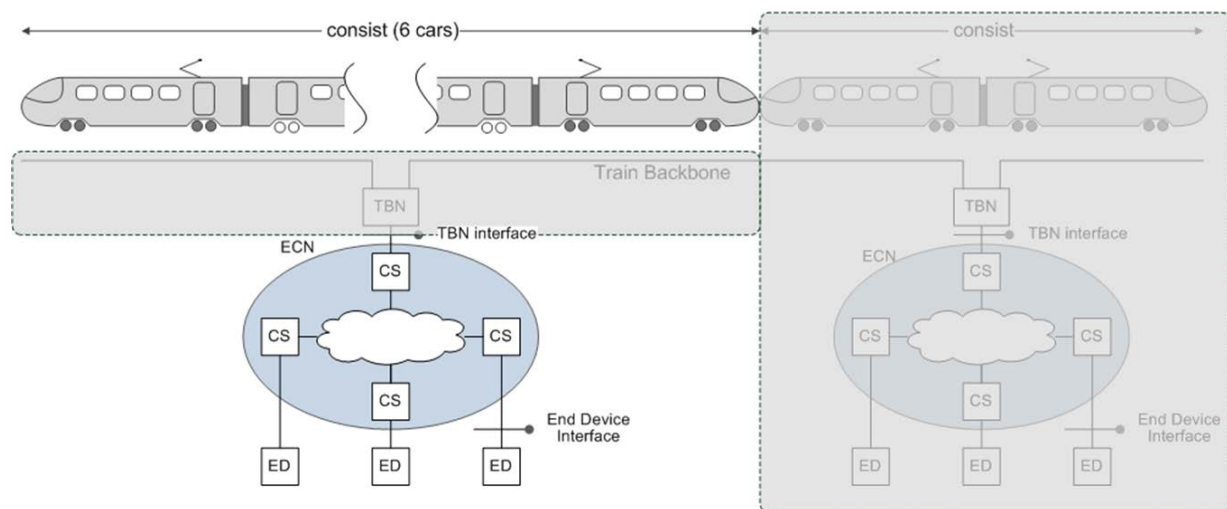


Figure 8-4 Train Backbone and Consist Networks for MRTJ

Figure 8-5 shows the network architecture of the TIS for MRTJ. End Device (ED) describes on-board equipment such as VVVF inverters, brake control units, and air conditioners. Gateways correspond to CCU/MON units and RIO units. Point-to-point interfaces such as RS-485 communication, digital I/O, etc. are placed in the sub-network level.

Note: The description above applies only between the RIO and EDs. Communication between EDs are out of scope for Toyo Denki.

Figure 8-5 shows the network configuration for MRTJ.

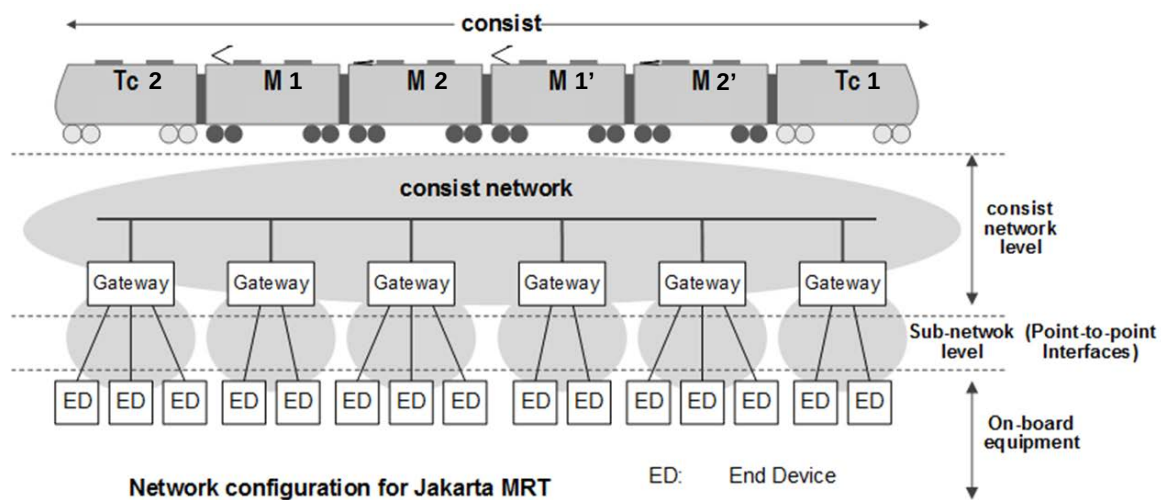


Figure 8-5 Network Configuration for MRTJ

8.1. RS-485 and High-Level Data Link Control Protocol

As explained in the previous section, point-to-point interfaces between RIOs and on-board devices are not included in the consist network. Therefore, they don't necessarily need to comply with the TCN standard.

Because PS Section 15 requires RS-485 for the PA system and the PIS, and because RS-485 High-Level Data Link Control (HDLC) communication is service-proven in Japan, RS-485, HDLC is used for the communication between the RIO and each device.

RS-485 is a standard defining the electrical characteristics of drivers and receivers for use in balanced digital multipoint systems. The standard is published by the Telecommunications Industry Association/Electronic Industries Alliance (TIA/EIA).

HDLC is a bit-oriented code-transparent synchronous data link layer protocol developed by the International Organization for Standardization (ISO).

The combination of RS-485 and HDLC protocol is a serial interface between the TIS and devices in the train. It has been used numerous times for TIS applications in Japan.

The relationship between RS-485 and HDLC is described below. Figure 8.1-1 shows the hierarchical structure of RS-485-based communication protocol. RS-485 or EIA-485 covers the physical layer and HDLC protocol covers the data-link layer.

The protocols of higher levels depend on the application, i.e. the transmission specification between the TIS and other devices.

Figure 8.1-1 shows an Open Systems Interconnection (OSI) 7-layer model.

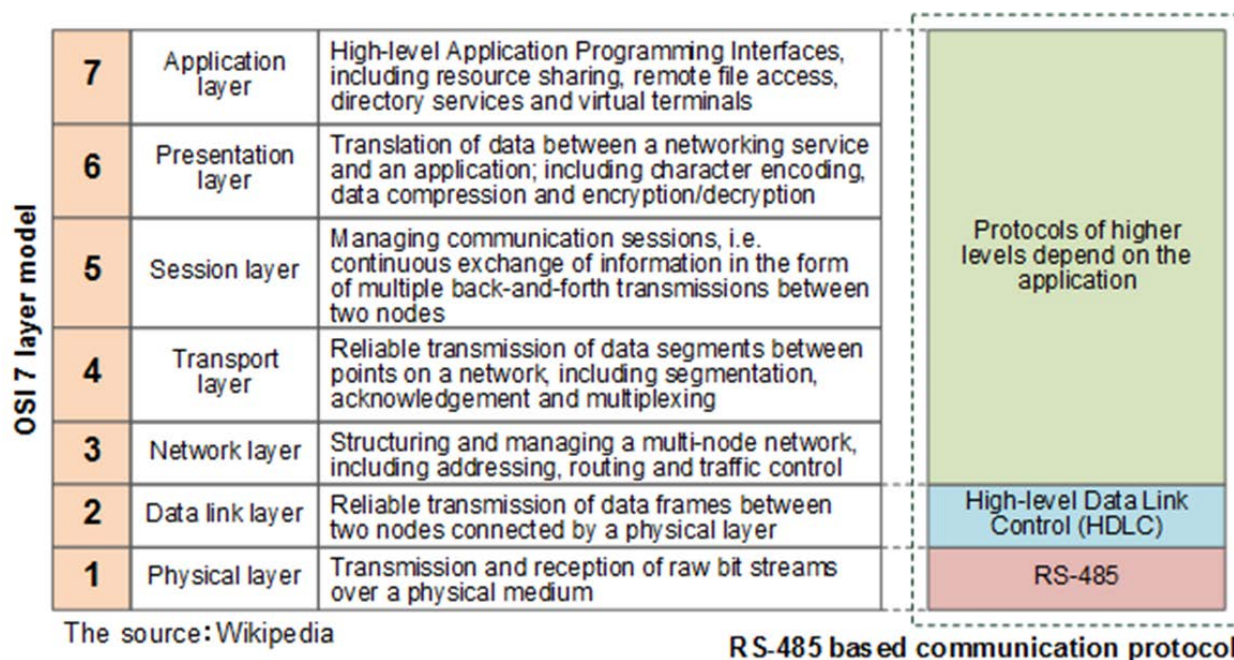


Figure 8.1-1 OSI 7-Layer Model

8.2. Variation of Consist Networks

Table 8.2-1 shows three variations of consist networks in IEC 61375. The TIS uses Ethernet because ECN has superior characteristics for transmission rates and maximum number of nodes. These characteristics are required to achieve the required advanced functions of the TIS.

Table 8.2-1 shows consist networks in accordance with IEC 61375 series.

Table 8.2-1 Consist Networks in Accordance with IEC 61375 Series

Item	MVB	CCN	ECN
(1) IEC61375 No.	3-1	3-3	3-4
(2) Transmission rate	1.5Mbps	500kbps, 1Mbps	100Mbps (10Mbps)
(3) Cable length	200m	100m (500kbps)	100m
		30m (1Mbps)	
(4) Channel coding	Manchester code	Bit stuffing	4B/5B MLT-3
(5) Maximum number of nodes	32	126	Practically unlimited

The proposed TIS is in accordance with the international standards. Accordingly, applied technologies are adopted from many train systems worldwide. Therefore, it is considered that this TIS is a well-proven system.

9. Components

9.1. CCU/MON Unit

A CCU/MON unit is installed in a leading vehicle. The CCU/MON unit consists of a Programmable Logic Control (PLC) unit, an Ethernet switch, a power supply unit, and a relay board. The PLC unit works as an Ethernet master of the control/monitoring transmission system.

Figure 9.1-1 shows the CCU/MON unit.

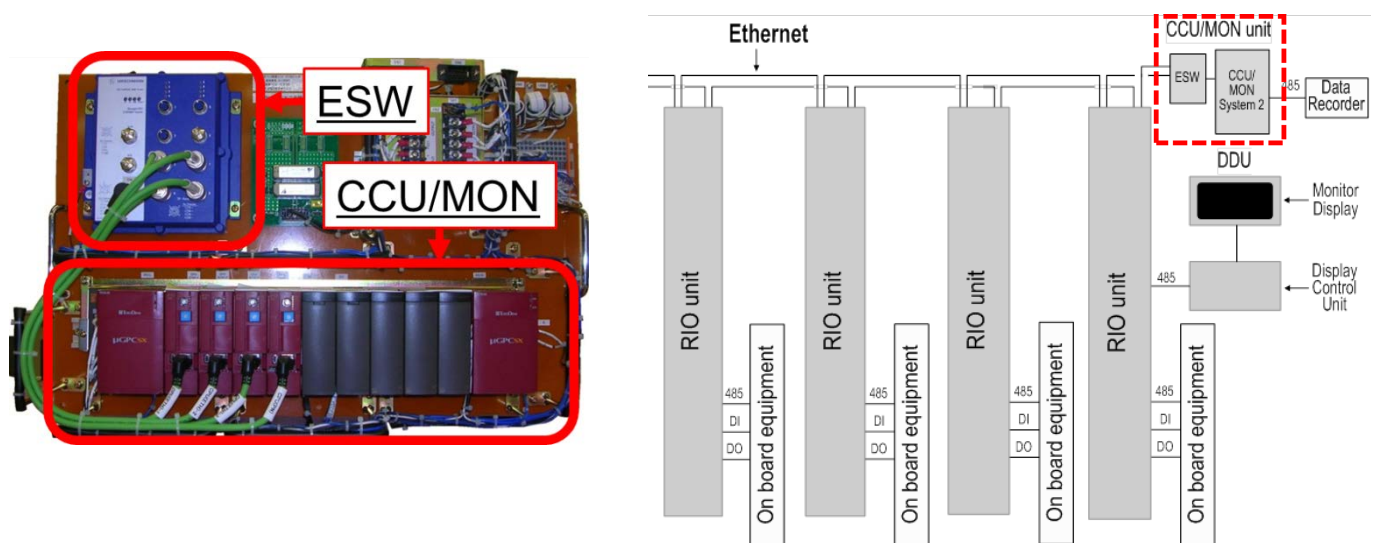


Figure 9.1-1 CCU/MON Unit

9.1.1. Function of Central Control Unit (CCU)

- Controls the control data transmission system.
- One CCU is installed as a part of the CCU/MON unit in each leading car.
- The CCU consists of a PLC unit, and works as an Ethernet master.

9.1.2. Function of Monitoring Unit (MON)

- Controls the monitoring transmission system.
- One MON is installed as a part of CCU/MON unit in each leading car.
- The MON consists of a PLC unit, and works as an Ethernet master.

9.1.3. Function of Ethernet Switch (ESW)

- Manages transmission of system traffic between cars and supports redundancy.

9.1.4. Role of CCU

- Communicates with an RIO on each car.
- Transmits operation commands from the ATO/ATP master controller to on-board equipment via RIOs.
- Transmits information required for train control to on-board equipment via RIOs.

9.1.5. Role of MON

- Communicates with an RIO on each car.
- Receives information to be displayed from on-board equipment via RIOs.
- Manages status data, fault data, and diagnostic data of on-board equipment, and records them.
- Transmits operation data to the Data Recorders.

9.1.6. Role of ESW

- An ESW is installed as a part of CCU/MON unit in each leading car.
- The ESW is connected with RIOs by Ethernet to consist ECN with the ring topology.
- The ESW changeovers communication route when a current route fails.

9.1.7. Characteristics of CCU/MON Unit

- Conforms to IEC61131.
- High performance PLCs are used.
- Complies with CP108 PS Volume 4 Section16.9.

9.2. RIO Unit

An RIO unit is installed in each car. The RIO unit is a rack-shaped unit containing several PCBs. Each RIO unit contains:

- One power supply board
- Two CPU boards
- Multiple Interface boards.

The two CPU boards have Ethernet interfaces. One connects to CCU/MON System 1, and the other connects to CCU/MON System 2.

The Interface boards include:

- RS-485 communication boards.
- Digital input/output boards.

The input/output circuits of the interface boards are designed as redundant systems.

The interface boards are configured to cope with the variation of input/output signals from different car types.

Figure 9.2-1 shows an RIO unit.

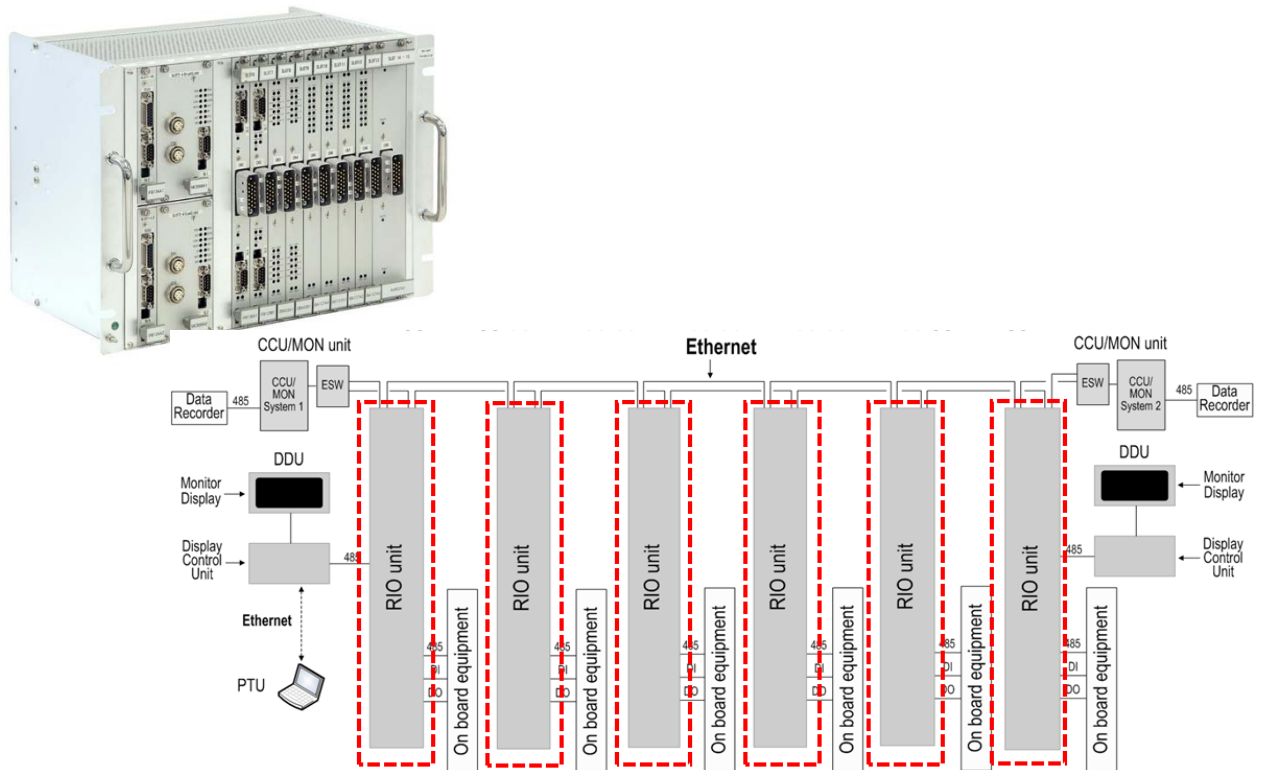


Figure 9.2-1 RIO Unit

9.2.1. Function of RIO

- Interface with each on-board equipment item.
- RS-485, Digital inputs/outputs.
- Communication interface with CCU/MON.

9.2.2. Role of RIO

- Communicates with the CCU and the MON by Ethernet.
- Has interfaces of digital signals and RS-485 communications.
- Transmits operation commands from the CCU to on-board equipment.
- Transmits data from on-board equipment to the MON.

9.2.3. Characteristic of RIO

- Main CPU board and power supply are installed for each communication system.

9.3. Driver's Display Unit

A DDU is installed on the cab console of both leading cars. Each DDU consists of following equipment:

- Monitor Display
- Display Controller
- Cables between the display and controller.

The DDU display screen indicates the status information of various devices on the network. The operator can touch the display to control devices such as lighting and air conditioning.

The Monitor Display consists of:

- 10.4-inch color LCD unit
- Touchscreen
- Control module
- Brightness control knob
- Buzzer volume adjustment.

The Display Controller consists of:

- Power supply module
- CPU board
- RS-485 communication module.

The Display Controller controls the Monitor Display and the Touchscreen. The Display Controller communicates with the RIO units via an RS-485 interface. The CPU board has no mechanically movable components, such as a hard disk or cooling fan. The CPU board runs the Windows-embedded operating system, which supports both English and Indonesian. No shutdown operation is necessary when the Display Controller's power supply is turned off.

Figure 9.3-1 shows the Monitor Display and Figure 9.3-2 shows the Display Control Unit.

Generally, the LCD requires more frequent maintenance than the display control circuits. Therefore, the DDU consists of two units so that it is easy for maintenance personnel to replace only the monitor display.

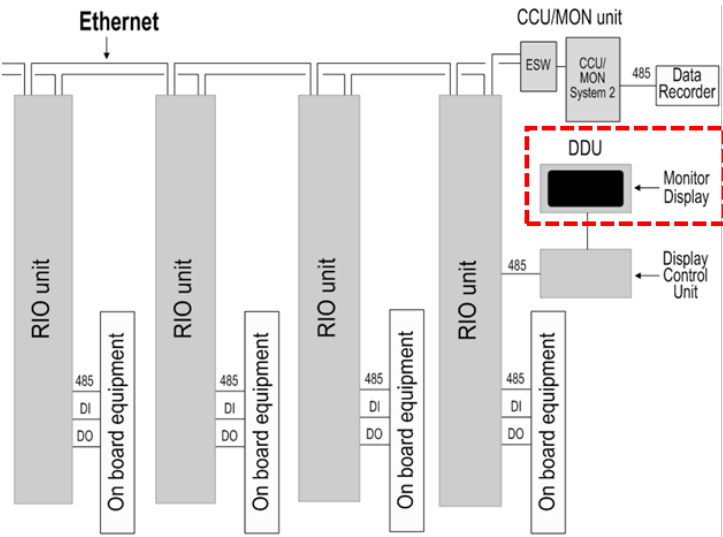


Figure 9.3-1 Monitor Display

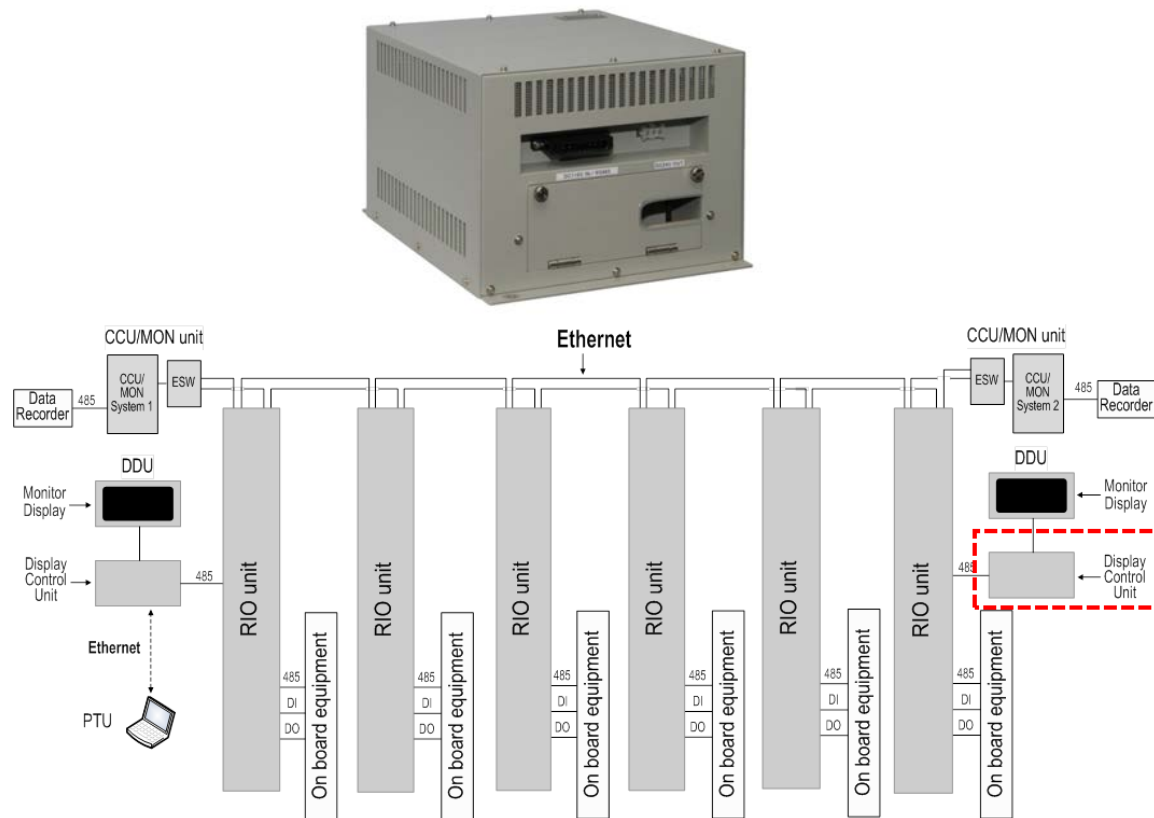


Figure 9.3-2 Display Control Unit

9.3.1. Specifications of Monitor Display

General specifications of the monitor display are shown in Table 9.3.1-1.

Table 9.3.1-1 General Specifications of Monitor Display

Item	Specification
Input Voltage	DC 24 V $\pm$ 5%
Input Current	Max. 1 A
Ambient Temperature	0 to 50 deg. C
Storage Temperature	-20 to 60 deg. C
Humidity	30%RH to 85%RH (without condensation)
Touchscreen	Resistive Touchscreen
MTBF of Monitor Control Unit	Approx. 1.5×10^4 hours

9.3.2. Function of Monitor Display

- Control of equipment by screen operation.
- Provide screen indication of status of each equipment unit on the train.
- A monitor display is installed as a part of DDU in each Tc cab.

9.3.3. Role of Monitor Display

- Displays status data and fault data of the train.
- Changes indications on the display.
- Sets operation parameters on the display.
- Performs sequential tests.

9.3.4. Characteristic of Monitor Display

- Display device: TFT color LCD.
- Display operation: Touch panel.
- Display size: 10.4 inches.
- Resolution: 800 x 600 dots.
- Backlighting life: 50,000 hours.

9.3.5. Characteristics of Display Control Unit

Table 9.3.5-1 shows the general specifications of the Display Control Unit.

Table 9.3.5-1 General Specifications of Display Control Unit

Item	Specification
Rated Input Voltage	DC 110 V
Output Voltage	DC 24 V $\pm$ 5%
MTBF of Monitor Control Unit	Approx. 2.8×10^7 hours

The display control unit is not a redundant unit, but it has high reliability. Based on the MTBF of the unit (2.8×10^7 hours), MTBF per train for the display control unit is estimated at 1.4×10^7 hours. This MTBF value has the same order as the MTBF value of ECN shown in Table 10.1-1 (2.44×10^7 hours). Therefore, the display control unit does not degrade reliability of the TIS significantly, even though the unit is not redundant.

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9.3.6. Function of Display Control Unit

- A Display Control Unit is installed as a part of the DDU in each Tc cab.

9.3.7. Role of Display Control Unit

- Display Control Unit has interfaces between the RIO and Monitor Display to transmit data to be displayed and set parameters
- Controls indication and touch panel
- When communication errors occur, the RIO detects them.

9.3.8. Characteristic of Display Control Unit

- OS: Windows 7 (Note: The DDU is never connected to the Internet.)
- Data output: USB port.

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10. Network Topology for TIS

10.1. Network Topology

Figure 10.1-1 shows the network topology (Ring topology).

As a characteristic of TIS, Ethernet Consist Network (ECN), a ring configuration with a high level of redundancy is applied as the base.(Employer's Requirements CP108 Vol.4 section16.4)

-Transmission path in normal operation

■ :Sending and Receiving Ports

Red Dotted Line : System 1

Yellow Dotted Line : System2

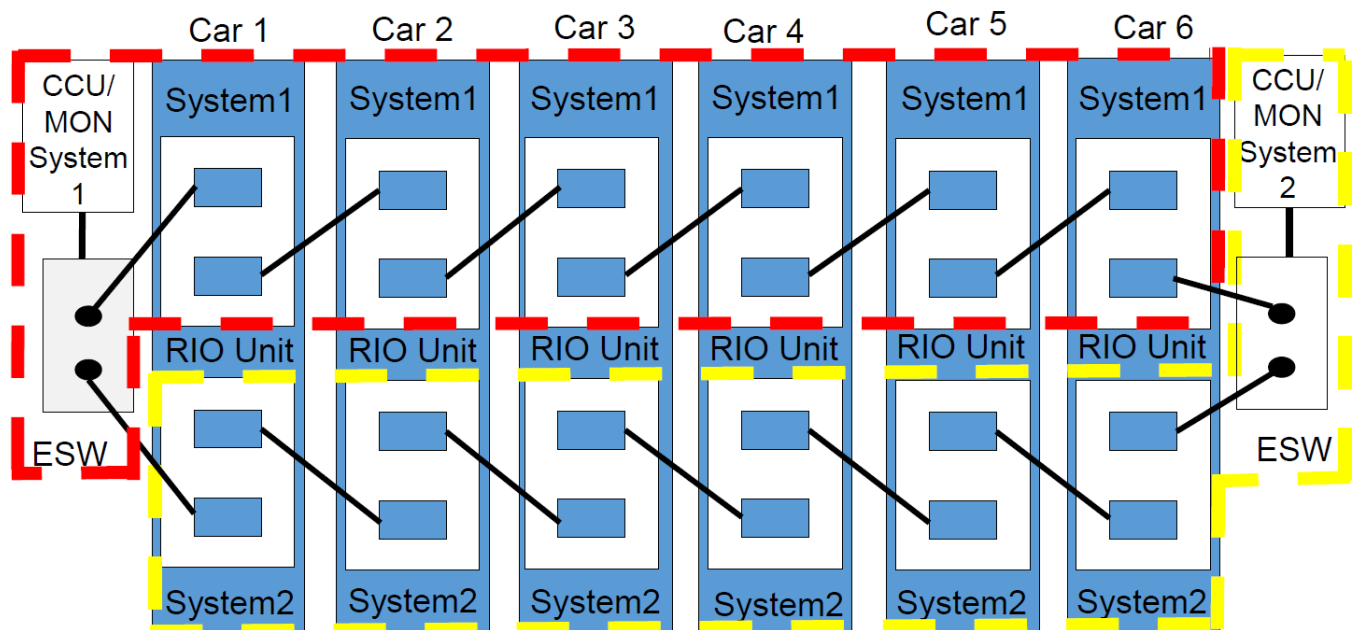


Figure 10.1-1 Proposed Network Topology (Ring Topology)

Figure 10.1-2 shows the consist network communication under normal conditions.

Transmission path of system 1 is used for explanation. When in normal operation, information will be transmitted to each car as shown by the blue dotted arrow.

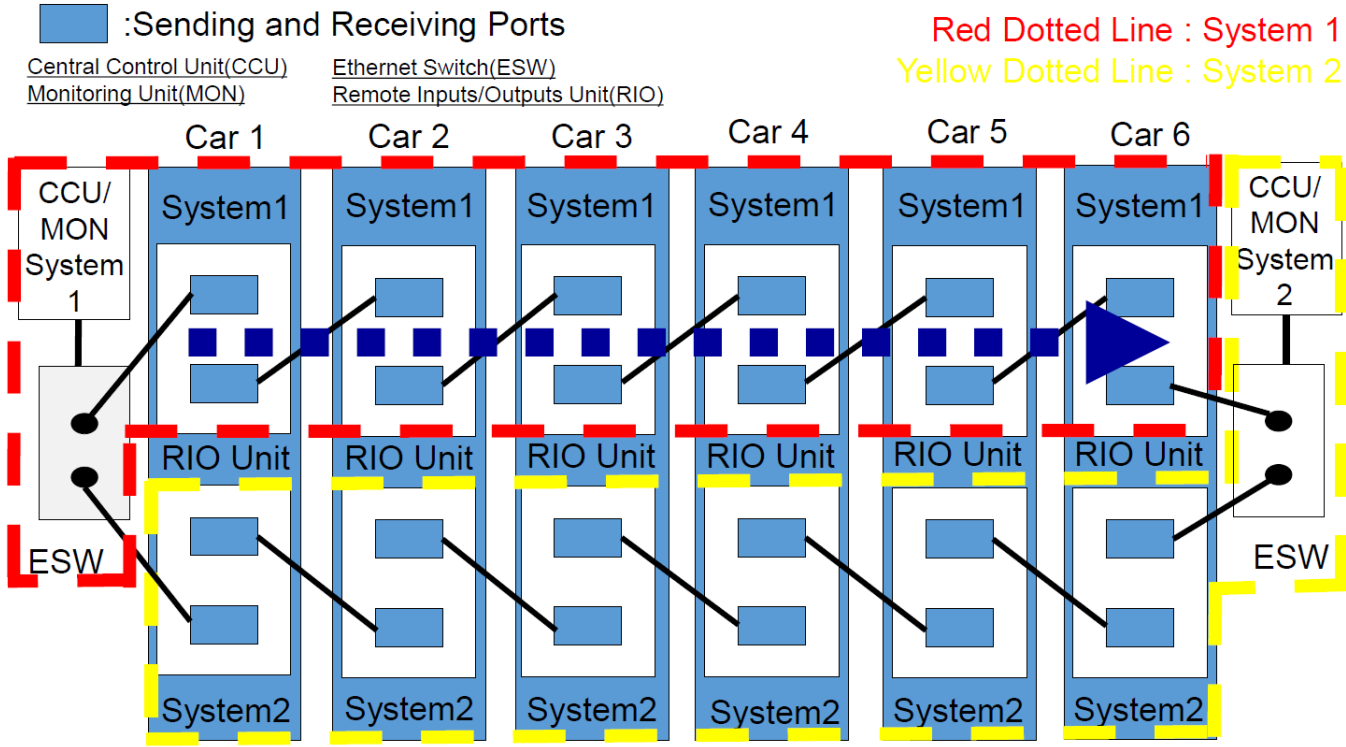


Figure 10.1-2 Consist Network Communication Under Normal Conditions

Figure 10.1-3 shows the consist network communication under failed conditions on a cable.

If a transmission path is blocked, the transmission path is rerouted to avoid the fault location, as shown in the blue dotted arrow, thereby realizing redundancy.

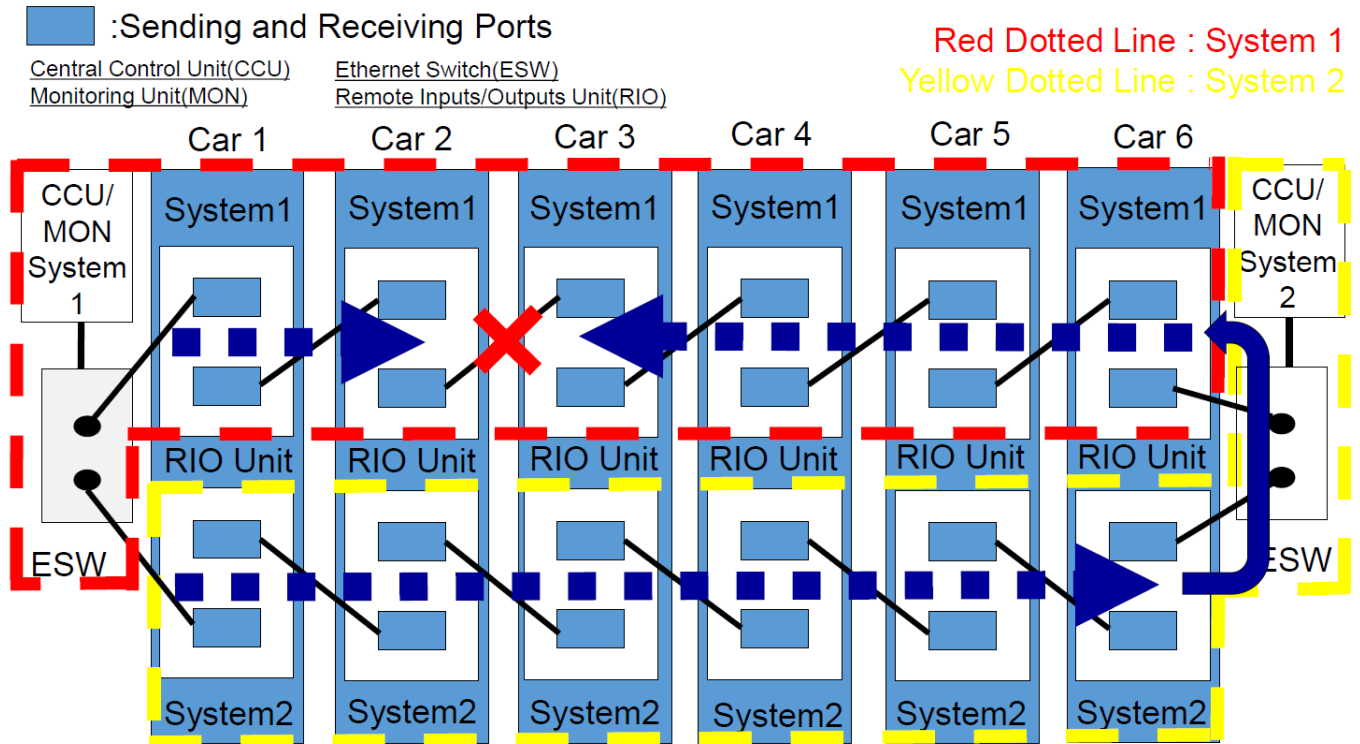


Figure 10.1-3 Consist Network Communication Under Failed Conditions on a Cable

Figure 10.1-4 shows the consist network communication under failed conditions on a RIO unit.

When a fault occurs in RIO Unit (System 1), the affected system is switched to System 2.

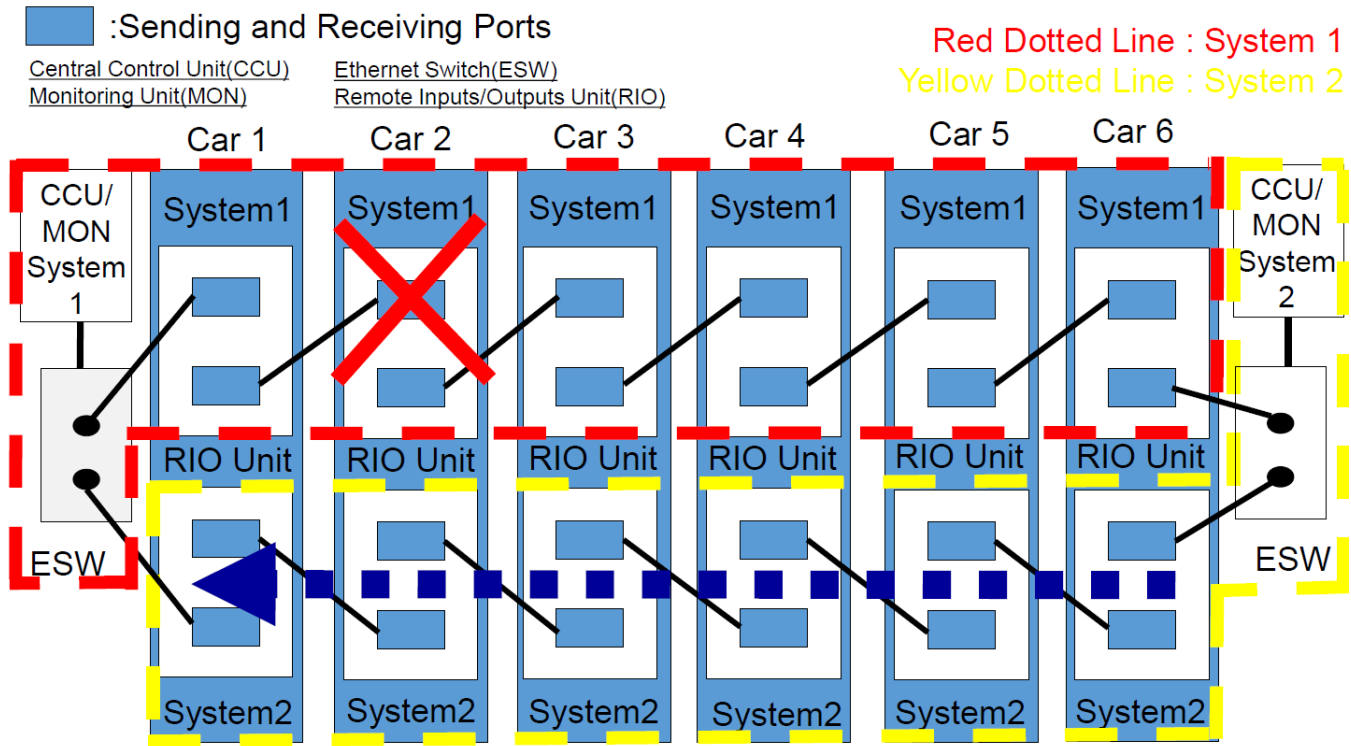


Figure 10.1-4 Consist Network Communication Under Failed Conditions on a RIO

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Table 10.1-1 shows the list of network topologies.

Table 10.1-1 List of Network Topologies

Item	Description	Note
The number of ports required	2 (both sides)	—
Complexity level of the bypass cont	It is possible to form the only bypass route to change the transfer direction to the adjacent node.	—
Transmission speed	100Mbps	—
Transmission delay	10^{-6} to 10^{-4} sec.	—
MTBF of network	2.44×10^7 (h)	Without common cause failure.
Number of Failures	Approx. 1.64	Per 16 fleets and 15 years 19 hours/day, 365 days/year

The failure rate of the network affects the failure rate of the TIS.

The failure rate of the network shown in Table 10.1-1 is the following:

- Number of Failures is approx. 0.007 per fleet per year.

The failure rate of the TIS is the following based on our field data:

- FIT: approx. 3400
- Number of Failures is approx. 0.57 per fleet per year.

The data in Table 10.1-1 shows that the failure rate of the network has a very low impact on the failure rate of the TIS.

The redundancy of TIS is shown as follows:

- Complies with PS 16.9.
- A single-point failure of any individual part of the TIS shall not result in a loss of data.
- The performance impact of any single-point failure of the TIS is minimized.
- If a transmission path is blocked by a wire breakage or other such failure, ring configuration is used to reroute the transmission path to avoid the location that is blocked.
- If a RIO Unit fails, the TIS switches to the functionally redundant unit and performs the transmission.
- The performance impact is minimized so that a single-point failure does not result in a loss of data.

11. Specifications of TIS

11.1. General Specifications

Table 11.1-1 shows the general specifications of the TIS.

Table 11.1-1 General Specifications of TIS

No.	Items	Specification		
1	Ambient temperature	CCU/MON unit RIO unit	Starting temperature	-10 to +45 deg. Celsius
			Operating temperature	-20 to +65 deg. Celsius
			Storage temperature	-25 to +70 deg. Celsius
		DDU (Monitor display, display controller)	Starting temperature	0 to +60 deg. Celsius
			Operating temperature	-20 to +60 deg. Celsius
			Storage temperature	-25 to +60 deg. Celsius
2	Relative humidity	Guaranteed humidity for performance: 90% or less (without dew drop)		
3	Anti-vibration	Standard: JIS E 4031		
4	Power source	110VDC (70VDC ~ 121VDC)		
5	Withstand voltage	1500VAC, 50/60 Hz, for 1 min		

11.1.1. Quantity of Devices in Trainset

Table 11.1.1-1 shows the quantity of TIS devices in a trainset.

Table 11.1.1-1 Quantity of Devices in Trainset

No.	Unit Name	Quantity						Total Quantity
		Tc1	M1	M2	M1'	M2'	Tc2	
1	CCU/MON Unit	1	-	-	-	-	1	2
2	RIO Unit 1 (for Tc Car)	1	-	-	-	-	1	2
3	RIO Unit 2 (for Tc Car)	1	-	-	-	-	1	2
4	RIO Unit (for M1 Car)	-	1	-	1	-	-	2
5	RIO Unit (for M2 Car)	-	-	1	-	1	-	2
6	DDU (Monitor display + Display controller)	1	-	-	-	-	1	2
7	Accessories	Connectors for CCU/MON units, RIO units, and DDUs						1 set

11.2. Specifications of CCU/MON Unit, RIO Unit, DDU, and Wiring

11.2.1. CCU/MON Unit TA1692-A•B - M

Table 11.2.1-1 shows the specifications of the CCU/MON Unit TA1692-A-B-M.

Table 11.2.1-1 Specifications of CCU/MON Unit TA1692-A•B - M

No.	PCB Name	Quantity	Notes
1	CPU module	3	PLC CPU module
2	Ethernet communication module	1	Ethernet I/O controller
3	Digital Input module	1	24VDC, photo-coupler insulation
4	Digital Output module	1	24VDC, photo-coupler insulation
5	Serial communication module	1	One Asynchronous communication
6	Power Supply module	2	110VDC input (24VDC output)
7	Ethernet switch	1	Number of TX Ports- 8
8	Relay board	1	TIS Abnormality output board
9	Power Supply Unit	1	—

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11.2.2. RIO Unit1 TA1695-A-M (for Tc Car)

Table 11.2.2-1 shows the specifications of the RIO Unit 1 TA1695-A-M (for Tc Car).

Table 11.2.2-1 Specifications of RIO Unit 1 TA1695-A-M (for Tc Car)

No.	PCB Name	Quantity	Notes
1	MCB119 Main CPU board (for the control/monitoring transmission)	2	Ethernet has one receive terminal and one transmit terminal respectively embedded.
2	IBA137 Digital input board	5	110VDC, photo-coupler insulation 4ch-1com×2, 2ch-1com×1
3	OBA59 Digital output board	1	110Vdc Photo-MOS relay insulation 1ch-1com×8
4	IFB159 RS485 communication board	2	Four channels, HDLC or Asynchronous communication is selectable for each channel
5	PW99 Power Supply	1	110VDC input

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11.2.3. RIO Unit 2 TA1695-B-M (for Tc Car)

Table 11.2.3-1 shows the specifications of RIO Unit 2 TA1695-B-M (for Tc Car).

Table 11.2.3-1 Specification of RIO Unit2 TA1695-B-M (for Tc Car)

No.	PCB Name	Quantity	Notes
1	MCB119 Main CPU board (for the control/monitoring transmission)	2	Ethernet has one receive terminal and one transmit terminal respectively embedded.
2	IBA137 Digital input board	5	110VDC, photo-coupler insulation 4ch-1com×2, 2ch-1com×1
3	OBA59 Digital output board	1	110VDC Photo-MOS relay insulation 1ch-1com×8
4	IFB159 RS485 communication board	2	Four channels, HDLC or Asynchronous communication is selectable for each channel
5	PW99 Power Supply	1	110VDC input

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11.2.4. RIO Unit TA1695-C-M (for M1 Car)

Table 11.2.4-1 shows the specifications of RIO Unit TA1695-C-M (for M1 Car).

Table 11.2.4-1 Specifications of RIO Unit TA1695-C-M (for M1 Car)

No.	PCB Name	Quantity	Notes
1	MCB119 Main CPU board (for the control/monitoring transmission)	2	Ethernet has one receive terminal and one transmit terminal respectively embedded.
2	IBA137 Digital input board	4	110VDC, photo-coupler insulation 4ch-1com×2, 2ch-1com×1
3	OBA59 Digital output board	1	110VDC, Photo-MOS relay insulation 1ch-1com×8
4	IFB159 RS485 communication board	3	Four channels, HDLC or Asynchronous communication is selectable for each channel
5	PW99 Power Supply	1	110VDC input

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11.2.5. RIO Unit TA1695-D-M (for M2 Car)

Table 11.2.5-1 shows the specifications of RIO unit TA1695-D-M (for M2 Car).

Table 11.2.5-1 Specifications of RIO Unit TA1695-D-M (for M2 Car)

No.	PCB Name	Quantity	Notes
1	MCB119 Main CPU board (for the control/monitoring transmission)	2	Ethernet has one receive terminal and one transmit terminal respectively embedded.
2	IBA137 Digital input board	4	110VDC, photo-coupler insulation 4ch-1com×2, 2ch-1com×1
3	OBA59 Digital output board	1	110VDC, Photo-MOS relay insulation 1ch-1com×8
4	IFB159 RS485 communication board	2	Four channels, HDLC or Asynchronous communication is selectable for each channel
5	PW99 Power Supply	1	110VDC input

11.2.6. Driver's Display Unit

11.2.6.1. Monitor Display TA1696-A-M

Table 11.2.6.1-1 shows the specifications of monitor display TA1696-A-M.

Table 11.2.6.1-1 Specifications of Monitor Display TA1696-A-M

Item	Specification	
Display	TFT color LCD/10.4 inches	
Touch panel	Resistance film system	
Number of pixels	800 (H) × 600 (V) dots	
Color	16M colors (260,000 colors)	
Interface	VGA	1 channel
	RS232C	1 channel (for touch panel control)

11.2.7. Display Control Unit TA1697-A-M

Table 11.2.7-1 shows the specifications of display control unit TA1697-A-M.

Table 11.2.7-1 Specifications of Display Control Unit TA1697-A-M

Item	Specification	
CPU	Intel ATOM Processor E3826 Dual Core/1.46GHz	
System memory	4GB	
Flash disk	SSD 16GB	
Interfaces	RS-485	1 channel (for the communication with the RIO unit)
	RS-232C	1 channel (for touch panel control)
	VGA	1 channel
	USB	2 channels (for maintenance by the manufacturer)
	LAN	2 channels (for maintenance by the manufacturer)
OS	Windows Embedded Standard	

11.2.8. Portable Test Unit

Table 11.2.8-1 shows the specifications of the PTU.

Table 11.2.8-1 Specifications of PTU

Item	Specification
OS	Windows 7
CPU	Intel ® CORE
Memory	DDR3L-SDRAM 4GB
Hard disk	SSD 256GB
Interfaces	LAN (for the communication with CCU unit) and others

Note: The operation manual will describe that a PC for the PTU is never connected with the Internet.

12. Specifications of DDU

12.1. Transition Diagram

Figure 12.1-1 shows the screen transition diagram of the TIS display.

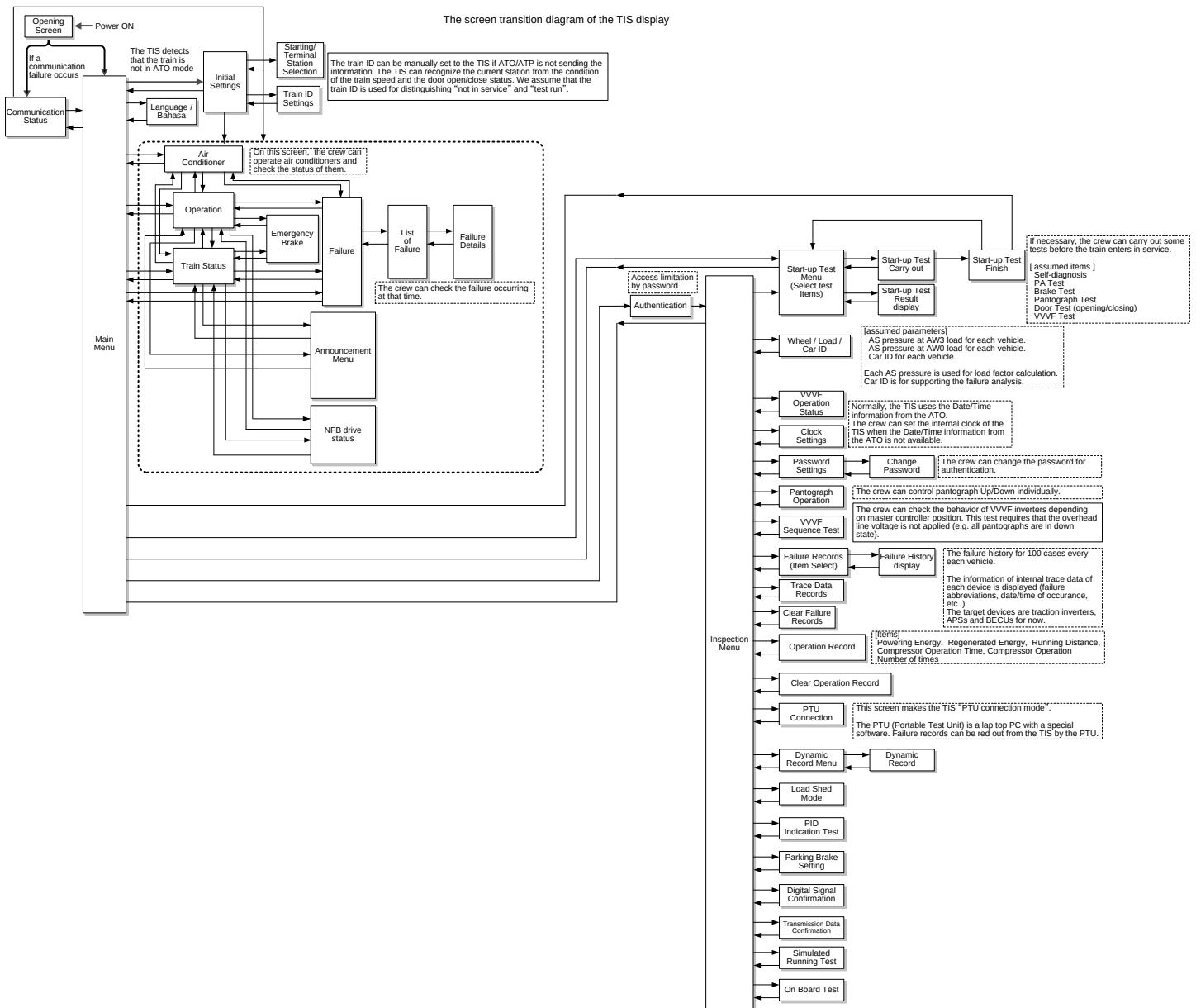


Figure 12.1-1 Transition Diagram of TIS Display

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12.2. Specifications of Screen Indications

Refer to CDR for TIS (SRR-RST-ELC-0024) to show the specification of screen indications.

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Attachment-1

Check Sheet of Technical Design Report

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No.	Item	Check
1	The main body of TDRs should contain all the necessary drawings, calculations, and explanatory notes, so that MRTJ/JMCMC/DGR can basically understand the contents by reading main body only.	✓
2	Structure of main body should be the same with the structure of the Particular Specification of the Employer's Requirement (PS).	✓
3	Calculation result and the like should be described in the main body of TDR, referenced document for such calculation result have to be clearly mentioned there.	—
4	Applicable Standards shall follow priority specified in the Particular Specifications of the Employer's Requirement (PS). In case EN, IEC or other non-specified Standards is used, reasons to apply such Standard has to be described.	✓
5	TDRs should contain the compliance sheet to all the clauses in PS after chapter 4 GENERAL REQUIREMENTS.	✓
6	TDRs should contain "Evidences of Compliance" to PS for any important items.	—
7	TDRs should contain the list of applicable Standards and corresponding items in the Chapter 3 "Codes and Standards" of each TDR.	✓
8	TDRs should confirm all the applicable codes & standard including DGR Regulation.	✓
9	TDRs should describe compliance to the concerned DGR for each TDR.	✓
10	Specification for main equipment/parts should be described in the main body of the TDRs.	✓
11	TDRs should describe maintenance system in the main body.	✓
12	Title and numbering for figures and tables in all TDRs have to be put on.	✓

Approved: M. Saito Checked: M. Azuma Planned: K. Nakamura

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Attachment-2

Reply Sheet from Employer's Representative Comments

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No.	Page	Clause	Comment & Questionnaire	Response
1	-	-	PS 16.1: SC shall consider providing sample pictures of TIS display at the Mock-up review.	Added explanation in section 6.
2	-	-	PS 16.1: SC shall consider whether Indonesian language can be displayed in TIS display or not.	Added explanation in section 6.
3	-	-	PS 16.1: Because only one Driver Display Unit (DDU) shall be equipped in a front end (Tc) car, how the redundancy can be secured at the time of DDU failure should be described in the main body of the TDR. Regarding display of data relating to ATP/ATO system, SC shall discuss and confirm purpose/function/necessity of each data with CP107 Contractor. For example, SC should consider what can be displayed in TIS display when speedometer and/or indicator lamp are out of order.	Added explanation in section 6.
4	-	-	PS 16.1: It should be described in the main body of the TDR that Speed display device shall obtain data from which device. For example a tachometer generator shall not be attached to the 1st axle of a Tc car because braking effort may have an impact on it. Or countermeasure should be described when the displayed speed is different from the speed recognized by ATP/ATO.	Added explanation in section 6.
5	-	-	PS 16.1: Process of the Self inspection done by TIS should be described. For example whether the back end car shall be inspected after the front end car is inspected. Daily inspection for ATP/ATO shall be described too.	Added explanation in section 6.7
6	-	-	PS 16.2 (c): Detailed explanation on "Driver Display Unit" should be described in the main body of the TDR.	Added explanation in section 9.3
7	-	-	PS 16.2 (g): Control of compressor (equalizing operation hour) and control of air conditioner with the extended power supply should be described.	Added explanation in section 6.

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No.	Page	Clause	Comment & Questionnaire	Response
8	-	-	In case there's any difficulty in complying with Appendix 8 in the General Specification of the Employer's Requirement, SC shall consult with MRTJ/ JMCMC regarding countermeasure for such difficulty.	We will consult with MRTJ/ JMCMC if necessary.

No.	Page	Clause	Comment & Questionnaire	Response
1	-	-	PS4.1.2, 16.2 and 16.3: SC should consider how the speed information can be obtained in case of ATP/ATO cutoff or failure. SC should check with CP107 if it is possible to provide speed information with ATP/ATO's NFB off. SC should consider whether the other equipment, such as BCU can output speed information.	SC will add explanation in Functional Explanation of Control Circuit (SRR-RST-ELC-0006).
2	-	-	PS16.2(a): SC should add explanation of "Self inspection before putting into service"	Added explanations in Section 6.7.
3	-	-	PS16.1: TIS controls subsystem operation and avoiding huge inrush currents for APS is typical example. SC should explain how TIS performs (or does not perform) the operation coordination between subsystems. Major inductive loads for APS are air compressor and air conditioning and SC should explain how simultaneous starting of such equipment is prevented.	Added an explanation in Section 7.
4	-	-	Section6: The English and Indonesian display of DDU should be able to be switched rather than displayed together.	Added explanations in CDR for TIS (SRR-RST-ELC-0024).
5	-	-	Section6: SC should add explanation of Figure 6.8-1 in the main body. SC should also add pictures of TIS display in Section 6 for easy understanding.	Added explanations in CDR for TIS (SRR-RST-ELC-0024).
6	-	-	Section9: SC should add whether the touch panel is optical or electrostatic.	Added the explanation in Table 9.3-1.
7	-	-	SC should show examples of TIS screen including Indonesian editions at mock-up review.	SC will show an example of TIS screen including an Indonesian edition at mock-up review.

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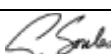
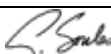
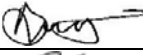
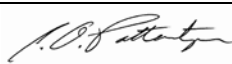
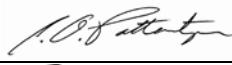
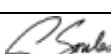
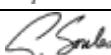
No.	Page	Clause	Comment & Questionnaire	Response
1	-	-	SC shall explain how system redundancy can be assured using 1 DDU in each cab. Section 9.3 should contain DDU reliability data.	Revised Section 9.3.
2	-	-	SC shall explain how TIS can obtain train speed information in detail, including the cases of ATP/ATO onboard failed or cutoff.	Refer to Functional Explanation of Control Circuit (SRR-RST-ELC-0006)
3	-	-	SC shall rectify and explain how the train can be operated when TIS is not in operation including failure case. SC shall confirm the consistency of explanations between the TDRs of TIS and control circuit.	Refer to Functional Explanation of Control Circuit (SRR-RST-ELC-0006)

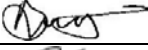
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No.	Page	Clause	Comment & Questionnaire	Response
1	-	-	AMENDMENT SHEET: Listed items for this revision are not sufficient.	Modified the AMENDMENT SHEET for this revision.
2	-	-	Many miss descriptions of Table and Figure numbers in main statements.	They were modified.
3	-	-	Section 5.3: Specification of 'enclosure' is not given.	The word 'enclosure' was changed to 'electric equipment box'. The referent of specification was added in Section 5.3.
4	-	-	Section 6.4.2: TOR for data recorder describes data recorder uses CF card.	The Data Recorder uses CF Card. The response is in the comment.
5	-	-	Section 6.8: Screen examples should be updated reflecting the design meetings.	The screen examples are updated and some screen examples are attached in CDR.
6	-	-	Section 7, Figure 7-1: No abbreviation for EPFC.	The abbreviation for EPFC is added in Figure 7-1.
7	-	-	Section 7: Description for timer relay should be corrected.	The description is modified.
8	-	-	Section 7 .1: No abbreviation for PCC.	The acronym "PCC" was not used.
9	-	-	Section 9.2: Explanation for RIO should be updated. No enough explanation was given how 'ring topology' can be implemented by using the RIO.	The explanation was modified.
10	-	-	Many miss descriptions for superscript (especially in Section 10).	They were modified.
11	-	-	Section 10.3.5: No longer needed.	It was deleted.
12	-	-	Section 12.3 and 12.4: TBD and X should be fixed.	It was modified and moved to Section 11.
13	-	-	Section 15, table 15-1: Marking of 'X' should be changed to 'v'.	Table 15-1 was modified and moved to CDR.
14	-	-	SC should confirm consistency of the contents with the other related TDRs.	It was confirmed.
15	-	-	SC should confirm corresponding CDR contains referred information. Document reference number should be added when the other document(s) would be referred.	It was confirmed.

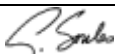
(Filename): TDR for TIS (Rev.I).docx	Rev. I
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Design Verification Checklist

IVF Review		Signature	Date
NS Preparer	[Name] Kousuke Nakamura		29th April, 2016 23rd May, 2016 19th July, 2016 21st July, 2016 26th July, 2016 9th September, 2016 12th September, 2016 28th October, 2016 23rd March, 2017 15 th April, 2017
SC Independent Internal Department Reviewer	[Name] Hiroshi KOBAYASHI		29th April, 2016 23rd May, 2016 19th July, 2016 21st July, 2016 26th July, 2016 9th September, 2016 12th September, 2016 28th October, 2016 23rd March, 2017 15 th April, 2017
IVF Reviewer	Rev. Nil - George Soules (GS)		3 rd May, 2016
	Rev. A - George Soules (GS)		26 th May, 2016
	Rev. B – Withdrawn, not reviewed	N/A	N/A
	Rev. C - Andy Pattantyus (IAP)		22 nd July, 2016
	Rev. D - Andy Pattantyus (IAP)		26 th July, 2016
	Rev. E - George Soules (GS)		10 th September, 2016
	Rev. F - George Soules (GS)		12 th September, 2016
	Rev. G - George Soules (GS)		22 nd December, 2016
	Rev. H - George Soules (GS)		6 th April, 2017
	Rev. I – Bryan Lawrence		20 th April, 2017
IVF Disposition	Rev. Nil - Fail		3 rd May, 2016
	Rev. A - Pass		26 th May, 2016
	Rev. B – Withdrawn, not reviewed	N/A	N/A
	Rev. C - Fail		22 nd July, 2016
	Rev. D - Pass		26 th July, 2016
	Rev. E – Fail See remedial action #27		10 th September, 2016
	Rev. F - Pass		12 th September, 2016
	Rev. G - Fail		22 nd December, 2016

	Rev. H - Fail		6 th April, 2017
	Rev. I - Pass		20 th April, 2017
If Fail, IVF suggest remedial action.	1. Update the documentation to explicitly list every subsystem the TIS interfaces with per PS 16.1. 2. Update the documentation to state that fault information will appear on a diagram showing the whole train and each car per PS 16.2(c). 3. Update the documentation to explicitly list all TIS maintenance support functions per PS 16.2(d). 4. Update the documentation to explicitly list all TIS passenger service functions per PS 16.2(e). 5. Update the documentation to explicitly list all TIS operation support functions per PS 16.2(g). 6. Update the documentation to state that the TIS shall be protected from EMI generated by on-board power equipment per PS 16.5. 7. Update the documentation to state that a data download facility shall be provided per PS 16.7. 8. Update the documentation to state that TIS programming shall allow for easy data entry and function changing per PS 16.8.	 George Soules	3 rd May, 2016
	9. See Evidence of Compliance table and correct all items marked with an "X" 10. Revise TDR Sections 7 and 8 to describe and specify TIS connections to the suspension system. 11. Revise TDR Sections 7 and 8 to describe and specify TIS connections to the lighting system. 12. Revise TDR Section 6.3 to	 Andy Pattantyus	22 nd July, 2016

	describe and specify TIS compliance to PS 16.2(g). 13. Revise TDR Section 6 to describe and specify how TIS acquires the crew number. 14. Revise TDR Section 6 to describe and specify how TIS transmits data to wayside systems through the train radio system. 15. Revise TDR Section 6 to describe and specify how TIS continuously acquires and records data. 16. Revise TDR to describe and specify the TIS programming environment, and to describe how the TIS complies with PS 16.8. 17. Revise TDR to describe and specify how TIS software structure complies with IEC 61131. 18. Revise TDR Table 3-1 by adding IEC 61375-2 and IEC 61375-3, 19. Revise TDR to describe how TIS complies with JIS C 0920 (2003). 20. Revise TDR to describe how TIS complies with JIS E 5006 (2005). 21. Revise TDR to describe how TIS software structure complies with IEC 61131-1 (2013). 22. Revise TDR Table 3-1 by adding IEC 61375-3. 23. Revise TDR to specify and describe how TIS Reliability, Availability, Maintainability and Safety complies with IEC 62278 (2002). 24. Revise TDR to specify and describe how TIS equipment complies with IEC 62279 (2015). 25. Revise TDR to specify and		
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	describe how TIS equipment complies with IEC 62425 (2007). 26. Revise TDR to specify and describe how TIS equipment complies with IEC 62497-1 (2010).		
	27. Compliance Matrix Table 4-1 for PS 16.2(b) refers to Section 6.8.3, but that section does not address the PS requirement: "Operation status display such as train speed, OCS voltage, traction motor voltage/current, auxiliary power voltage / current, battery voltage, air pressure, etc." Section 6.8.3 and Figure 6.8.3-1 describe Operation Display, but do not describe or show voltages or air pressure. Revise the Compliance Matrix to indicate how the TDR complies with PS 16.2(b).		10 th September, 2016
	28. See comments in TDR.		22 nd December, 2016
	29. See comments in TDR		6 th April, 2017
If Fail, NS Response	[Describe actual remedial action] 1. A subsystem list has been added to Section 6. 2. Section 5 has been revised. 3. Section 5.1 has been added to Section 5 as a subsection. 4. Section 5.2 has been added to Section 5 as a subsection. 5. Section 5.3 has been added to Section 5 as a subsection. 6. Section 4.3 has been revised. 7. Section 5 has been revised. 8. Section 5.1 has been added to Section 5 as a subsection.	[Name and signature]  Masashi Azuma	24 th May, 2016
	9. We cannot find any items marked with an "X" on the Compliance Table. 10. A description has been added on the note under the Table 7-1. 11. The Table 7-1 has been	[Name and signature]  Masashi Azuma	26 th July, 2016

	revised. 12. The descriptions on Section 6.3 have been revised. 13. The descriptions on Section 6.3 have been revised. 14. The descriptions on Section 6.3 have been revised. 15. It has been described in Section 6.4.3 and 6.4.4. 16. It has been described in Section 6.4. 17. Section 5.3 has been revised. 18. The Table 3-1 has been revised. 19. Section 5.3 has been revised. 20. Section 5.3 has been revised. 21. Section 5.3 has been revised. 22. Section 5.3 has been revised. 23. Section 5.3 has been revised. 24. Section 5.3 has been revised. 25. IEC 62425 has been deleted from the Table 3-1 because this standard is not referred by the GS nor the PS. 26. Section 5.3 has been revised. The units are tested JIS E 5006 which is equivalent to IEC 60571. Regarding insulation for electronic units, IEC 60571 is equivalent to IEC 62497-1. Therefore, JIS E 5006 is enough to satisfy with IEC 62497-1.		
	27. In Section 4-1, P.S.16.2 (b), modified "Section 6.8.3" to "Section 6.8.4" in the column of "TDR Chapter Number". Modified the explanation in Section 6.8.4.	[Name and signature]  Masashi Azuma	12 th September, 2016
	28. Modified document according to IVF and ER's comments.	[Name and signature]  Masashi Azuma	23 rd March, 2017

	29. Modified TDR according to IVF's comments.	[Name and signature]  Masashi Azuma	15 th April, 2017
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General	Description	Comments	Completed By (Initials)
Scope of Design (Package)	TIS		
Subject No. (Doc)	SRR-RST-ELC-0008I		
Doc/Dwg No.	BC161082	Inserted BC15**** from the DVT. Replace it with the actual internal doc no.	GS
Revision	Nil A B C D E F G H I		
Description	Technical Specification of TIS		
Number of sheets (or pages)	76 Pages		
Documentation to Support Verification			
PPT (calculations, justification)	PPT for TIS (0427) PPT for TIS (1023)		GS
System Function Description (SFD)			
Integration Matrix	Technical Specification of TIS		
Driving Requirement (Pick one)	List Requirement		
- GS/PS - Ind. MoT - JIS, JRIS - EN, IEC, etc. - Intl' Best Practice	P.S. Section.16 Minister of Transportation Regulation PM175 Year 2015 JIS C 0920 JIS E 5006 IEC 61131-1 IEC 61131-3 JIS E 4031 IEC 61375-1 IEC 61375-2 Series IEC 61375-3 Series IEC 62236-3-2 IEC 62278 IEC 62279		
Verification Method (singly or in combination)	List Verification Method(s)		
Similarity	Historical is shown in Technical	TDR section 10 and	GS

• Historical • Calculation/Drawings • Design Review (Meeting minutes)	Specification of TIS (SRR-RST-ELC-0008A) Design Review; CNT8.MTG.2016.06-0112	PPT for TIS (0427) section 9, show Experiences, but they don't indicate if the experience is with the Toyo Denki solution being proposed. Clarify how the experience is applicable to this project. Added the explanation in TDR section 10	
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DVCR Form - IVF's Evidence of Compliance Table

DVC Requirements	Requirement Description	IVF Review	Initials	Date
PS 16.1	<i>An open-system microprocessor-based TIS shall perform data acquisition processing, communication and display functions required to control car and train operation, and to detect, analyze and report on train operation status and subsystem failures including but not limited to:</i> (a) traction equipment; (b) braking equipment; (c) auxiliary power supply; (d) air compressor system; (e) air conditioning equipment; (f) suspension system; (g) battery and battery charging system; (h) saloon door system; (i) on-board ATP/ATO equipment; (j) on-board communication system; (k) lighting system (l) data recording system.	✓ TDR Table 7-1 Lists the Subsystems connected with the TIS, including: Propulsion Control Unit (Traction), Brake Control Unit, Auxiliary Power Supply, Air Compressor, Air Conditions, Battery, Battery Charger, Door Control Unit, ATP/ATO, PIDS, PA System, PEC, Train Radio, CCTV, and Data Recorder and lighting system. ✓ TDR Section 7.1 describes monitoring of lights. ✓ TDR Sections 7 and 8 specify connection to the Propulsion Control Unit. * TDR does not describe TIS connection to the suspension system. * TDR does not describe connection to the lighting system. ✓ Note under Table 7-1 describes how TIS connects to BECU to receive suspension system data.	IAP IAP	7/22/2016 7/26/2016

PS 16.2	<i>The functions of TIS shall include, but not be limited to;</i>			
	<i>(a) Self inspection before putting into service incorporated with self-diagnosing of major equipment and checking of door operation, train start and train stop.</i>	✓ TDR Section 6.7.4 describes the self-diagnostic functions. ✓ TDR Section 6.8 describes the self inspections and self-diagnostic functions of the Maintenance Display.	IAP	7/22/2016
	<i>(b) Operation status display such as train speed, OCS voltage, traction motor voltage/current, auxiliary power voltage/current, battery voltage, air pressure, etc.</i>	✓ TDR Section 6.8 describes the Operation Display screens.	IAP	7/22/2016
	<i>(c) 6.2.3 Fault display and instructions to Driver. Fault information shall be shown on an overview diagram that presents the whole train and shall be shown on an individual car basis.</i>	✓ TDR Section 6.1.2 describes the fault monitoring and display. Figure 6.1.2-1 shows an example of driver's display showing faults for the whole train, organized by individual car.	IAP	7/22/2016
	<i>(d) Maintenance support such as fault logging, accumulated travel distance, accumulated power consumption, etc.</i>	✓ TDR Section 6.4.5 describes accumulated running distance. ✓ TDR Section 6.4.7 describes the Accumulated Electric Power for the PCU. ✓ TDR Section 6.7.14 describes the collection of failure record.	IAP	7/22/2016
	<i>(e) Passenger service such as setting destination, saloon temperature and automatic PA, etc.</i>	✓ TDR Section 6.6.1 describes the setting of information for operational use, including end destination and stop stations. ✓ TDR Section 6.5.2 describes how the TIS controls the Passenger Information Display System (PIDS), including SDDs and EDDs. ✓ TDR Section 6.5.1 describes the air	IAP	7/22/2016

		conditioning equipment control. ✓ TDR Section 6.5.3 describes the automatic broadcasting system control.		
	(g) Operation support such as setting of train number and crew number through the ATP system, and transmitting data to wayside systems through the train radio system.	✓ TDR Section 6.6.1 describes how TIS receives train number from the ATO or the DDU. ✓ TDR Section 6.8 describes manual setting of train number. * TDR Section 6.3 does not contain the information showing compliance to PS 16.2(g). * TDR does not describe how TIS acquires the crew number. * TDR does not describe how the TIS transmits data to wayside systems through the train radio system. ✓ TDR Section 6.3 shows compliance to PS 16.2(g). ✓ TDR Section 6.3 describes how TIS acquires the crew number. ✓ TDR Section 6.3 describes interface with CP107 and the train radio system.	IAP IAP	7/22/2016 7/26/2016
PS 16.3	The TIS shall continuously record data relevant for analysis and investigation into causes of subsystem failure and train operating status immediately prior to failure.	* TDR does not describe how TIS continuously acquires and records data. ✓ Sections 6.4.3 and 6.4.4 describe how TIS continuously acquires and records data.	IAP IAP	7/22/2016 7/26/2016
PS 16.4	A single point failure of any individual part of the TIS shall not result in a loss of data. The performance impact of any single point failure of the TIS shall be minimized.	✓ Specified in Section 10.1.1. Fault tolerant architecture described in Sections 10.3.1 and 10.3.2 and conclusion stated in	IAP	7/21/2016

		Section 10.3.4.		
PS 16.5	<i>The TIS shall be immune to interference or performance degradation in the environment influenced by EMI and harmonics generated from the on-board power equipment.</i>	✓ Conclusion stated in Section 5.3, Qual. Tests in accordance with IEC 62236-3-2 assure immunity to EMI.	IAP	7/22/2016
PS 16.6	<i>The display unit shall be use TFT color LCD technology. Commands shall be entered through touch panel.</i>	✓ TDR Section 9.3 describes the DDU color display. ✓ TDR Section 9.3.3 specifies the TFT color LCD display. ✓ TDR Section 12.5 specifies the TFT color LCD display.	IAP	7/22/2016
PS 16.7	<i>A download facility shall be provided to download data and information from the TIS. The Contractor shall provide any software required to enable the data exchange between TIS and a laptop computer.</i>	✓ TDR Section 6.1.3 specifies a PTU for download by connection to the display control unit.	IAP	7/22/2016
PS 16.8	<i>The TIS programming shall allow for easy data entry and function changing and upgrading throughout the life of the system.</i>	*-TDR does not describe the TIS programming environment, and does not describe how the TIS complies with PS 16.8. ✓ TDR Section 6.4 describes the TIS parameter entry, function changing and data recording environment.	IAP IAP	7/22/2016 7/26/2016
PS 16.9	<i>The TIS shall be well-proven technology applied in rolling stock use. The software structure shall comply with IEC 61131: Programmable Controllers and communication network shall comply with IEC 61375-1: Electric railway equipment-Train bus Part 1. Train Communication Network or equivalent.</i>	*-TDR does not describe how TIS software structure complies with IEC 61131. ✓ TDR Section 5.3 specifies that the TIS software structure will comply with IEC 61131 series. ✓ TDR Section 8 specifies that the TCN and TIS satisfy IEC 61375-1 requirements.	IAP IAP	7/22/2016 7/26/2016
JIS C 0920 (2003)	Degrees of Protection Provided by Enclosure (IP Code)	*-TDR does not describe how TIS equipment complies with JIS C	IAP IAP	7/22/2016 7/26/2016

		0920 (2003) ✓ TDR Section 5.3 specifies that the TIS equipment will comply with JIS C 0920 (2003)		
JIS E 4031 (2013)	Rolling stock equipment - Shock and vibration tests Equipment	✓ TDR specifies that TIS equipment complies with JIS E 4031 (2013)	IAP	7/22/2016
JIS E 5006 (2005)	Electronic equipment for rolling stock	* TDR does not describe or specify that TIS equipment complies with JIS E 5006 (2005) ✓ TDR Section 5.3 specifies that TIS equipment will comply with JIS E 5006 (2005)	IAP IAP	7/22/2016 7/26/2016
IEC 61131-3 (2013)	Programmable Controllers – Part 3: Programming Languages	* TDR does not describe or specify TIS software structure compliance with IEC 61131-3. ✓ TDR Section 5.3 specifies that TIS software structure will comply with IEC 61131-3.	IAP IAP	7/22/2016 7/26/2016
IEC 61375-1 (2012)	Electronic railway equipment – Train communication network - Part 1: TCN Train Communication Network General Architecture	✓ TDR Section 8 specifies that the TCN and TIS satisfy IEC 61375-1 requirements. ✓ TDR Section 3 Table 3-1 lists IEC 61375-1.	IAP IAP	7/22/2016 7/26/2016
IEC 61375-2	Electronic railway equipment – Train communication network (TCN) - Part 2-1: Wire Train Bus (WTB) Electronic railway equipment – Train communication network (TCN) - Part 2-3: TCN communication profile. Electronic railway equipment – Train communication network (TCN) - Part 2-5: Ethernet Train Backbone.	✓ Although mentioned in Section 8, IEC 61375-2 is not applicable. ✓ TDR Section 5.2 states that TIS has no train backbone. ✓ TDR Section 5.3 states that Train Backbone is not required, since the train consist is a fixed 6-car train. ✓ TDR Section 8 specifies that the MRTJ vehicles are “closed trains” that have neither a train back bone nor a train backbone	IAP IAP	7/22/2016 7/26/2016

		node, satisfying IEC 61375-1 requirements. ✓ TDR Section 3 Table 3-1 lists IEC 61375-2.		
IEC 61375-3	Electronic railway equipment – Train communication network (TCN) - Part 3-1: Multifunction Vehicle Bus (MVB) Electronic railway equipment – Train communication network (TCN) - Part 3-3: CANopen Consist Network (CCN) Electronic railway equipment – Train communication network (TCN) - Part 3-4: Ethernet Consist Network (ECN)	✓ TDR Section 8 Table 8.2-1 describes consist networks that satisfy IEC 61375-3 requirements, and section 8.2 describes selection of ECN according to IEC 61375-3-4. ✓ TDR Section 10.3 describes the network topologies defined in IEC 61375-3-4. * TDR Section 3 Table 3-1 does not list IEC 61375-3. ✓ TDR Section 3 Table 3-1 lists IEC 61375-3.	IAP IAP	7/22/2016 7/26/2016
IEC 62236-3-2 (2008)	Railway applications – Electromagnetic compatibility – Part 3-2: Rolling stock Apparatus	✓ TDR Section 5.3 describes that qualification tests of the TIS are performed in accordance with IEC 62236-3-2. ✓ TDR Section 3 Table 3-1 lists IEC 62236-3-2.	IAP IAP	7/22/2016 7/26/2016
IEC 62278 (2002)	Railway applications – The specification and demonstration of Reliability, Availability, Maintainability, and Safety	* TDR does not describe or specify that TIS equipment complies with IEC 62278 ✓ TDR Section 3 Table 3-1 lists IEC 62278. ✓ TDR Section 5.3 specifies that the TIS and its software will be designed, manufactured and tested in accordance with IEC 62278 and IEC 62279.	IAP IAP	7/22/2016 7/26/2016
IEC 62279 (2015)	Railway applications – Communication signaling and processing systems – Software for railway control and protection	* TDR does not describe or specify that TIS equipment complies with IEC 62279	IAP IAP	7/22/2016 7/26/2016

	systems	✓ TDR Section 3 Table 3-1 lists IEC 62279. ✓ TDR Section 5.3 specifies that the TIS and its software will be designed, manufactured and tested in accordance with IEC 62278 and IEC 62279.		
IEC 62425 (2007)	Railway applications – Communication signaling and processing systems – Safety related electronic systems for signaling	*–TDR does not describe or specify that TIS equipment complies with IEC 62425. ✓ IEC 62425 has been deleted from the Table 3-1 because this standard is not referred by the GS nor the PS.	IAP IAP	7/22/2016 7/26/2016
IEC 62497-1 (2010)	Railway applications – Insulation – Part 1: Basic requirements – Clearances and creep age distances for all electrical and electronic equipment	*–TDR does not describe or specify that TIS equipment complies with IEC 62497-1. ✓ Section 5.3 has been revised. The units are tested JIS E 5006 which is equivalent to IEC 60571. Regarding insulation for electronic units, IEC 60571 is equivalent to IEC 62497-1. Therefore, JIS E 5006 is enough to satisfy with IEC 62497-1	IAP IAP	7/22/2016 7/26/106